International Financial Flows, Credit Allocation and Productivity

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Abstract

This paper examines how financial liberalization shapes domestic credit allocation and real economic outcomes using detailed bank-firm matched data for India. We exploit a 2004 reform that raised foreign shareholding limits in Indian banks and identify differences in credit allocation patterns across banks with varying exposure to foreign capital inflows. More-exposed banks received larger capital inflows, which relaxed their funding constraints and expanded their overall lending. This increase in bank lending was directed towards the more productive firms in the economy, raising their investment and reducing the overall dispersion of marginal revenue product of capital (MRPK) across firms. This improvement in credit allocation was associated with foreign equity inflows rather than foreign debt financing, highlighting the importance of the form of capital inflows. Quantifying the aggregate effects of this reallocation, we find that the liberalization generated significant gains in total factor productivity (TFP).

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1. Introduction

International financial flows play a key role in shaping economic activity, especially in emerging market economies. On the one hand, cross-border flows are an important source of external finance for domestic firms and banks — easing credit constraints, improving credit allocation, and spurring investment and productivity (Kose et al. (2008)). On the other hand, inflow surges are often associated with macroeconomic and financial stability risks — credit booms, lax lending standards, asset price bubbles, and even financial crises (Diaz-Alejandro (1985), Kaminsky and Reinhart (1999)). The ultimate effects of these financial flows may therefore depend on the form and channels through which they enter the recipient country (Ghosh et al. (2016), Ostry et al. (2010)). In this paper, we exploit a natural experiment afforded by the Indian government's 2004 decision to allow substantially greater foreign shareholding in Indian banks to examine what happens when inflows take the form of higher equity stakes in local financial intermediaries. We find that, far from resulting in lax lending standards, this liberalization was associated with better credit allocation and correspondingly, economically and statistically significant gains in aggregate productivity.

Early studies of capital inflows (Chari and Henry (2003), Henry (2000b), Henry (2000a)) mostly examined financial liberalization episodes in emerging markets using aggregate macroeconomic data in cross-country analyses. More recent work (Gopinath et al. (2017), Varela (2018), Bau and Matray (2023)) explores how liberalization and subsequent inflows affect misallocation and productivity. Much of this literature focuses on direct capital flows to firms. In emerging markets, however, few firms have direct access to international capital markets and foreign capital is typically intermediated through the domestic banking system. While existing studies have examined how inflows affect bank balance sheets and aggregate lending (Baskaya et al. (2017), Di Giovanni et al. (2017)), much less is known about how they affect credit allocation across firms and, in turn, real economic outcomes. In this paper, we seek to address this gap by examining the "bank-lending channel" of capital inflows — tracing how foreign equity flows affect banks' balance sheets, total lending, and credit allocation and thus firm performance and aggregate productivity.

Any such empirical analysis runs into three main challenges. First, capital flows are likely to be endogenous — attracted to countries that are expected to grow and to entities that are expected to be profitable. Accordingly, it is difficult to distinguish between the effect of the capital inflows and the conditions that attracted them in the first place. Second, an observed expansion in bank lending may reflect either an increase in credit supply by banks or greater credit demand by borrowers. Third, even when capital inflows increase aggregate lending, determining whether they are associated with improved credit allocation requires tracing credit across individual firm-bank relationships and connecting those changes to firm performance and aggregate productivity.

To address these challenges, we study a bank-equity liberalization episode in India. In March 2004, the Indian government raised the ceiling on foreign shareholding in domestic private banks from 49 percent to 74 percent of equity, reducing barriers to foreign investment and expanding banks' access to foreign capital. We exploit this change in foreign ownership restrictions as a natural experiment to examine the impact on individual banks' behavior, notably their credit allocation and the quality of their lending decisions.

Since, however, foreign investors' decisions in which banks to invest might be endogenous to these outcomes, we follow the literature (Paravisini et al. (2015), Mian and Sufi (2022), Cingano and Hassan (2022)) and rely on a predetermined measure of a bank's exposure to the reform to proxy for the post-reform flows. Specifically, we use a bank's pre-liberalization foreign shareholding as a measure of bank exposure to the reform on the grounds that banks with greater foreign ownership had stronger foreign connections and visibility to international investors, and were therefore better positioned to attract foreign capital once the ceiling was relaxed. Our identifying assumption is that the behavior of more exposed banks (for instance, in terms of lending decisions) did not differ significantly from that of less exposed banks in the pre-reform period, thus allowing us to infer that the change in behavior of the more exposed banks can be attributed to the post-reform foreign capital inflows [1]. We then combine bank balance sheet data with loan-level information on bank-firm credit relationships and firm-level balance sheet data and adopt a difference-in-differences methodology to trace the effects of the reform. Our analysis proceeds in three steps:

First, we examine how the reform affected banks' foreign ownership, funding structure and aggregate lending. We find that banks more exposed to the liberalization were able to raise more capital and borrow more from abroad in the post-reform period compared to less exposed banks with the differences being statisically significant. On the other side of the balance sheet, more-exposed banks increased their lending in the post-reform period by significantly more than less-exposed banks.

Second, using loan-level data, we isolate credit supply effects [2] and find that more-exposed banks increased their credit supply to firms significantly more than less-exposed banks in the post-reform period. This increase in credit supply was skewed towards firms that were more productive (i.e., have a high marginal revenue product of capital (MRPK)) and that were able to post more collateral with the difference again being statistically significant. Specifically, we find that a 1 percentage point increase in bank exposure to

[1] As elaborated in Section 2 and Appendix B below, several results justify this assumption. Bank exposure is largely unrelated to pre-reform characteristics. In addition, the data show no systematic differential trends prior to the reform and, in our regressions, we include bank fixed-effects to absorb any time-invariant and persistent differences in bank lending

[2] To isolate the impact of the reform on credit supply to firms, we use loan-level data and apply the Khwaja and Mian (2008) within-firm estimator, comparing changes in lending by differently exposed banks to the same firm in the same year. In complementary specifications, we use region-by-sector-by-year fixed effects to absorb local industry-specific demand conditions and isolate the impact of the liberalization on bank credit supply.

the liberalization was associated with a 1.7 percent increase in lending to more productive (high-MRPK) firms compared to a 1 percent increase to less productive (low-MRPK) firms. Moreover, while lending to both high- and low- productivity firms is increasing in their collateral (proxied by their fixed assets), the effect is stronger for high-MRPK firms.

Third, we show that more productive (high-MRPK) firms with greater exposure to the liberalization — in the sense that, pre-reform, a higher share of their credit was from banks that themselves had greater exposure — expanded their wage payments and investments in fixed assets in the post-reform period. (Conversely, we find a negative relationship for less productive (low-MRPK) firms). As the more productive firms expanded, the diminishing marginal product of capital meant that their MRPK fell by 0.8 percent while the contraction of the less productive firms raised their MRPK by about 1.2 percent (in response to a 1 percentage point increase in firm exposure). The narrowing of the MRPK differential between high- and low- productivity firms implied an increase in aggregate productivity (measured by the Solow Residual) of about 1.2 percent.

The results thus show that banks that were more exposed to the liberalization increased their lending and made better credit allocation decisions — both in terms of directing relatively more credit to productive firms along the intensive margin and in terms of forming new credit relationships. Regarding the latter, we find that, when more-exposed banks entered into new lending relationships in the post-reform period, they initially issued smaller loans and expanded their lending as the relationship matured and firms established their creditworthiness. More-exposed banks also increased their investment in human capital and experienced lower rates of loan restructuring, patterns consistent with improvements in banks' ability to screen and monitor borrowers.

We then examine why the reform may have been associated with these changes in bank behavior. The evidence points to the form of foreign capital as an important part of the mechanism: banks with greater post-reform foreign equity participation, and in particular larger foreign shareholder positions, experienced larger shifts in lending toward more productive firms, whereas we find no analogous relationship for foreign debt. These results are consistent with foreign equity affecting not only banks' funding constraints but also their incentives, governance, and internal decision-making, thereby contributing to the improvement in credit allocation.

Related Literature Our paper contributes to the long-standing debate over the benefits and risks of capital account liberalization. Financial integration can lower the cost of capital, relax financing constraints, and increase investment and growth (Henry, 2000b,a; Chari and Henry, 2003; Kose et al., 2008). At the same time, beginning with Diaz-Alejandro (1985)'s classic, *Goodbye Financial Repression, Hello Financial Crash* — a large literature has emphasized that financial liberalization and capital inflows can fuel rapid credit expansion and increase financial fragility (Kaminsky and Reinhart, 1999; Kaminsky and Schmukler,

2003). Further, Dell'Ariccia et al. (2012) show that credit booms can be accompanied by a deterioration in lending standards. An important theme in this literature is that the composition of flows matters: debt flows, particularly short-term and foreign-currency debt, tend to generate greater vulnerabilities than FDI and equity flows (Ostry et al., 2010; Ghosh et al., 2016). A notable qualification, however, is that foreign investment in the financial sector can itself contribute to vulnerability when it facilitates rapid domestic credit expansion (Ostry et al., 2010). This makes the Indian experience particularly interesting. The 2004 reform generated substantial foreign equity inflows into domestic banks and a large expansion in lending, yet we find that credit allocation improved rather than deteriorated. One possibility is that both the form and the sequencing of liberalization mattered: foreign capital entered banks as equity rather than short-term external debt, while foreign investors had already been able to establish minority positions in Indian banks before the ownership ceiling was relaxed further. Our results therefore suggest that rapid credit growth following financial liberalization need not come at the cost of poorer credit allocation, and that the type of capital entering the economy and the institutions through which it is intermediated are important for its effects.

Our paper relates most closely to Cingano and Hassan (2022), who use loan-level data from Italy to study how global capital inflows affect bank credit allocation and productivity. Like them, we use granular bank–firm credit data to connect international financial flows to allocative efficiency. Our setting differs in that we study a policy-driven liberalization of foreign equity ownership in banks in an emerging economy, rather than a broader increase in debt flows. We also relate to Baskaya et al. (2017), who show that capital inflows expand bank lending in Turkey; we show that foreign capital affects not only the quantity of bank credit but also its allocation across firms. Finally, our setting differs from the literature on foreign bank entry (Gormley, 2010; Detragiache et al., 2008; Chakraborty and Mitra, 2024). Rather than replacing domestic intermediation with foreign banks, the reform brought foreign capital into incumbent domestic banks. This distinction may be important because domestic banks retain their local information and lending relationships while gaining access to foreign capital and, potentially, improvements in governance and screening.

We also contribute to the literature linking international capital flows to misallocation and aggregate productivity (Bau and Matray (2023), Gopinath et al. (2017)). Bau and Matray (2023) show that relaxing foreign ownership restrictions in Indian manufacturing allowed firms to receive foreign investment directly, improving input allocation and productivity. We identify a distinct but complementary intermediary channel: foreign capital enters domestic banks and is reallocated across firms through bank lending. This channel is particularly relevant in emerging economies, where many firms cannot access international capital markets directly. By linking bank-level foreign inflows to loan-level credit allocation, firm-level input responses, and changes in the distribution of MRPK, we provide micro-founded evidence on how financial liberalization can affect aggregate productivity through

the domestic banking system.

In sum, while our paper is related to several strands of the literature, our unique setting — an exogenous, policy-driven inflow surge that took the form of both equity and debt flows, allows us to trace not only the impact of aggregate inflows but also how the form of inflows affects banks' lending decisions and the subsequent implications for firms' outcomes and aggregate productivity.

The rest of the paper is organized as follows. Section 2 describes the institutional setting, the 2004 banking liberalization, our identification strategy and the data. Section 3 examines the effects of the reform on bank balance sheets, including banks' funding and lending responses. Section 4 turns to the loan-level analysis and isolates the credit-supply effects of the reform. Section 5 examines the real effects on firms and quantifies the implications for aggregate productivity. Section 6 examines the mechanisms that led to an improvement in the allocation of credit by banks and Section 7 concludes.

2. Empirical Setting

2.1 Foreign Investment in Indian Banking and the Policy Reform

India operated as a closed, quasi-socialist economy until the late 1980s. A severe balance of payments crisis in 1991 paved the way for liberalization reforms, which led to the removal of several trade and licensing restrictions (Topalova and Khandelwal (2011)). Over time, there was also a gradual relaxation of foreign investment restrictions across various sectors, including the banking sector.

India's banking sector was predominantly state-owned until the early 1990s. In 1993–94, the government allowed new private banks to be established. However, both state-owned (public) banks and private banks remained subject to strict foreign shareholding limits of 20% of a bank's equity. In 2001, there was an attempt to liberalize the banking sector by increasing the cap on Foreign Direct Investment (FDI) in private banks to 49% [3]. Despite this policy change, the response of foreign capital flows to the banking sector was underwhelming. This was largely due to regulatory ambiguity over whether the 49% limit applied only to FDI or to total foreign shareholding, which also included Foreign Institutional Investment (FII), Foreign Portfolio Investment (FPI), holdings by non-resident Indians and other forms of foreign ownership. The lack of coordination between the Ministry of Finance, the Reserve Bank of India, and the Ministry of Commerce resulted in mixed messaging and considerable policy uncertainty [4]. Consequently, as shown in Figure 1, the

[3] DIPP, Press Note 4, 2001

[4] Excerpt from The Economic Times (Feb 17, 2002): "The Government, in May 2001, had announced FDI up to 49 per cent 'from all sources' in private sector banks. However, in the absence of any clarifications on whether this would include or exclude investments of FIIs, and no follow-up guidelines from the RBI, the old norms which capped FDI at 20 per cent, continued."

2001 policy announcement was not accompanied by any discernible change in the trajectory of foreign shareholding in Indian private banks. The regulatory ambiguity, combined with the 49% cap that prevented foreign entities from obtaining a majority shareholding, likely dampened investor interest and limited significant inflows into the banking sector.

The situation changed substantially in March 2004, when the government overhauled the policy governing foreign shareholding in Indian banks. The ceiling on total foreign shareholding in domestic private banks was increased to 74% of a bank's equity, while foreign shareholding in public banks continued to be capped at 20% [5]. In contrast to the 2001 reform, the new policy explicitly applied to total foreign ownership, which was communicated and implemented clearly, resolving much of the regulatory uncertainty surrounding foreign investment in the banking sector. The reform was followed by a sharp increase in foreign capital inflows into Indian private banks.

Interestingly, although the regulatory change applied specifically to private banks, public banks — which until then had almost negligible foreign shareholdings — also experienced an increase in foreign investment, albeit within their unchanged 20% regulatory limit. Thus, while the reform directly relaxed the foreign ownership constraint for private banks, the response of foreign investors was not confined to banks whose statutory limit increased. Figure 1 shows the sharp increase in foreign shareholding in Indian banks — both public and private — following the March 2004 reform. [6]

The influx of foreign capital took several forms, with Foreign Institutional Investors (FIIs) accounting for the largest share of investment. In addition, several banks attracted investments from institutions such as the International Finance Corporation, the Government of Singapore, and major international banks including HSBC, Rabobank, and Citicorp. Many banks also accessed international capital markets by issuing American Depository Receipts (ADRs) and Global Depository Receipts (GDRs) on overseas stock exchanges.

Overall, the 2004 liberalization of the banking sector proved effective in facilitating substantial foreign investment into Indian banks. It also enabled foreign investors to establish a more direct presence in the governance of several banks, including through representation on their boards.

2.2 Data

Our analysis draws on five main data sources. We use loan-level data on bank-firm lending relationships from the Ministry of Corporate Affairs (MCA), firm balance sheet data from

[5] DIPP, Press Note 2, 2004

[6] To formally test for a break in the trend of foreign shareholding around the reform date, we conduct a Chow test for a structural break at March 2004. The test rejects the null of no structural break for the full sample of all banks ($\chi^2(2) = 23.72$, $p < 0.001$), as well as separately for private banks ($\chi^2(2) = 14.73$, $p < 0.001$) and public banks ($\chi^2(2) = 15.08$, $p < 0.001$), confirming that foreign shareholding in both groups of banks exhibits a statistically significant break coinciding with the reform.

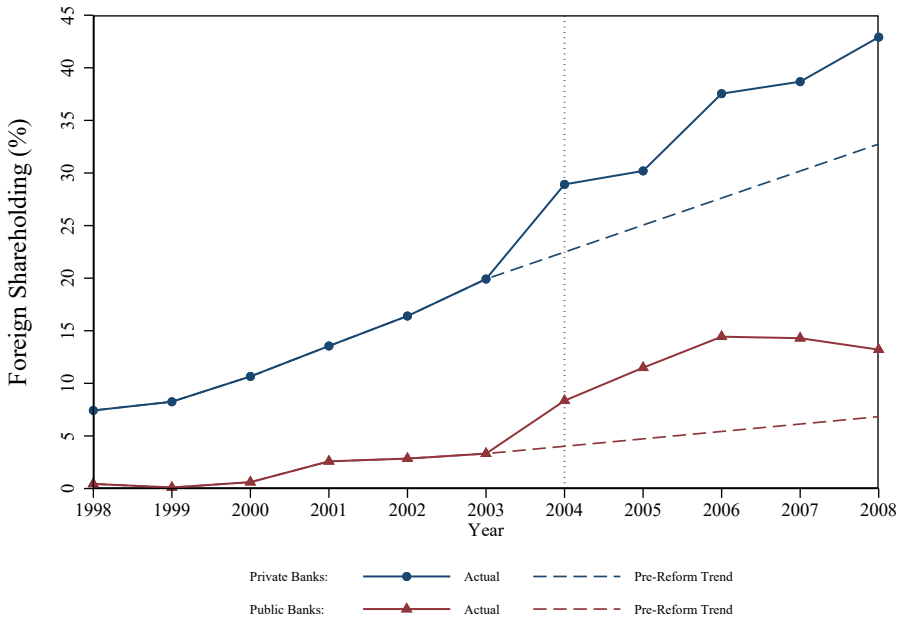


FIGURE 1. Foreign Shareholding in Indian Banks

Note: This figure plots the evolution of foreign shareholding (in %) for private and public banks in India. Solid lines show actual average foreign shareholding; dashed lines show the pre-reform trend projection, obtained by fitting a linear trend on the 1998–2003 period and extrapolating forward. The vertical dotted line marks the 2004 reform, which raised the foreign shareholding ceiling for private banks.

CMIE ProwessDx, bank balance sheet and other bank-level information from the Reserve Bank of India (RBI) and CMIE ProwessDx , foreign shareholding data from the Bombay Stock Exchange (BSE), and foreign equity flows data from the National Securities Depository Limited (NSDL).

Our primary source for bank-firm lending relationships is the MCA database, which has been widely used in recent work on financial-sector and bankruptcy reforms in India (Chari et al. (2021), Chopra et al. (2021), Kulkarni (2020)). We obtain this data for all firms for which balance sheet information is available in CMIE ProwessDx. The MCA data reports the loan amount, the date of origination, dates of modification and repayment, the names of the borrower and lender, and the address of the lending institution. Lenders include banks, non-bank financial institutions, development financial institutions, state governments, and, in some cases, individuals. We restrict the sample to loans originated by banks between 1998 and 2008. Because the data records both origination and repayment dates, we use them to construct outstanding credit for each firm-bank pair in each year. Details on the construction of the firm-bank panel are provided in Appendix A.

We match the MCA loan data with firm balance sheet information from CMIE ProwessDx, which contains detailed information on firm characteristics and outcomes. We also match

the loan data with bank balance sheet information and other bank-level variables from CMIE ProwessDx and the RBI. We separately compile detailed information on foreign shareholding in banks from archived webpages of the BSE.[7] The BSE data includes total foreign shareholding, holdings by different categories of foreign investors and the identities of foreign entities holding more than 1% of a bank's equity. Official series on sectoral FDI and FII flows in India are available only from 2012 onward and therefore do not cover our sample period. We instead use equity transaction data from NSDL to construct bank-level FII inflows over the period of analysis. Details on the construction of this measure are provided in Appendix A. Together, these data allows us to study the effects of the banking-sector liberalization on bank funding, credit supply, firm outcomes and aggregate productivity.

Table A1 presents summary statistics for the banks and firms in our sample. We restrict the bank sample to institutions that were listed prior to the liberalization, for which we can consistently measure foreign shareholding and exposure to the reform. This leaves a sample of 31 banks. We focus on the period 1998–2008 to avoid confounding the analysis with the effects of the global financial crisis. The firm sample contains about 15,000 firms on average, although the exact number of firms varies across specifications.

2.3 Identification

To study the impact of the liberalization on bank balance sheets and credit allocation, we use a difference-in-differences (D-i-D) framework that exploits cross-bank variation in exposure to the reform. As discussed above, following the reform, foreign investment increased in both private and public banks, even though the regulatory ceiling for public banks remained unchanged. We therefore include both public and private banks in the analysis.

Following the literature on bank liquidity shocks (Mian and Sufi (2022), Paravisini et al. (2015)), we measure a bank's exposure to the equity liberalization using its ex-ante foreign shareholding (specifically, FII and FDI holdings). The idea behind this measure is that banks with greater foreign investor presence prior to the liberalization were more likely to receive foreign capital once restrictions were relaxed. These banks were already better known to foreign investors and had stronger existing connections to international capital markets.

We assess the relevance of the exposure measure by examining whether higher pre-liberalization foreign shareholding predicts greater foreign equity inflows after the reform. Figure 2 plots the relationship between bank exposure and post-liberalization foreign equity inflows. The relationship is positive and statistically significant: banks with greater pre-reform foreign ownership did indeed receive larger foreign equity inflows after liberal-

[7] The CMIE ProwessDx database provides information on foreign shareholding only from 2001 onwards. We compile data on foreign shareholding for years prior to 2001 from the archived webpages of the BSE.

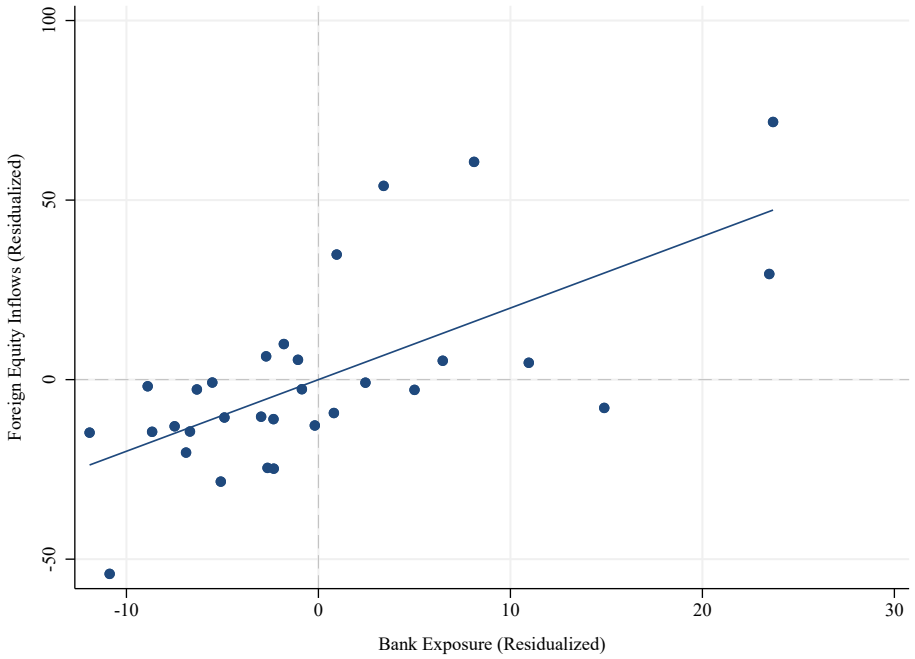


FIGURE 2. Bank Exposure and Share of Foreign Equity Inflows (Post-Liberalization)

Note: This figure plots the conditional correlation between bank exposure (measured using banks' pre-reform foreign shareholding (in %)) and post-liberalization gross foreign capital inflows (measured as a share of a bank's pre-reform assets). The correlation controls for key ex-ante bank characteristics such as bank assets, liquidity and capital ratio. The relationship is positive and significant — the exposure measure predicts post-reform foreign equity inflows into banks (coef: 1.994, SE = 0.486, t = 4.10, F-statistic = 16.84, R^2 = 0.4)

ization. [8] This provides support for using pre-reform foreign shareholding as a measure of banks' exposure to the reform.

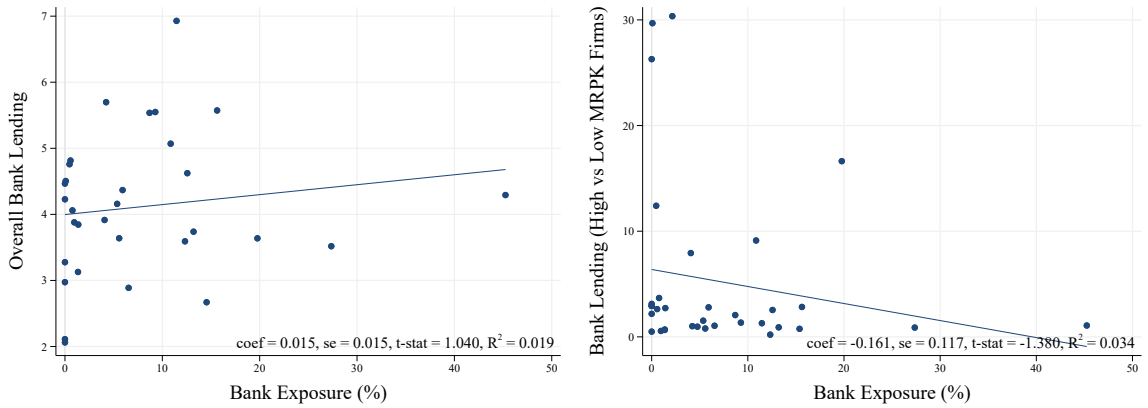
Threats to Identificaton — A key concern with the exposure measure is that banks with higher pre-liberalization foreign shareholding may have been systematically different from less exposed banks — larger, better capitalized, or better managed — and may therefore have followed different credit allocation trajectories even absent the reform. Appendix Figure B1A examines this by relating the exposure measure to a broad set of ex-ante bank characteristics, including size, liquidity, capitalization, profitability, asset quality, and foreign liabilities. Most characteristics are not significantly related to exposure; the only exception is the foreign liabilities share. [9] Overall, the balance tests provide little evidence that more exposed banks differed systematically from less exposed banks along these observable dimensions. [10] Since our key outcomes of interest are bank lending and credit allocation, we also examine whether exposure was systematically related to banks' pre-reform lending behavior. Figure 3 shows no significant relationship between bank exposure

[8] The positive relationship holds both unconditionally and conditional on bank-characteristics

[9] The correlation with ex-ante foreign liabilities share is expected given our definition of exposure.

[10] Further, the absence of pre-trends in Figure 5 implies that pre-reform exposure was uncorrelated with the outcome variables of interest, namely, banks' lending and credit allocation decisions.

and either pre-reform overall lending or the allocation of lending between high- and low-MRPK firms, providing additional evidence that more exposed banks were not already different in the outcomes that are central to our analysis.



(A) Overall Bank Lending (B) Bank Lending to High vs. Low MRPK Firms

FIGURE 3. Bank Exposure and Pre-Reform Lending Behavior

Notes: This figure plots the correlation between bank exposure and pre-reform overall bank lending (Figure A) and bank lending according to ex-ante firm productivity (Figure B). There is no significant correlation between bank exposure and pre-reform bank lending characteristics.

There are several other potential threats to the identification strategy. The first is the earlier increase in the FDI ceiling in 2001, which could complicate the interpretation of the pre-reform period. As discussed in the institutional background, however, this policy change was accompanied by considerable regulatory ambiguity and was not associated with a discernible change in the trajectory of foreign shareholding. A second concern is that public and private banks differ substantially in ownership and governance. Because foreign investment increased in both groups following the reform, we include both in the analysis and rely on continuous variation in ex-ante exposure rather than treating public banks as an untreated control group. The empirical specifications account for persistent differences across banks using bank fixed effects and by controlling for observable ex-ante bank characteristics. Finally, the period under consideration coincides with a broader expansion in capital inflows and credit in India. We account for this by using year fixed effects which control for changes in aggregate financial conditions that affected all banks. We discuss these concerns, along with additional tests of the identifying assumptions, in greater detail in Appendix B.

With the institutional setting, data and identification strategy in place, we now turn to the empirical analysis. We proceed in three steps. We first examine how greater exposure to the liberalization affected banks' balance sheets, focusing on their funding and aggregate lending responses (Section 3). We then use the loan-level data to isolate changes in credit

supply and study how banks reallocated credit across firms (Section 4). Finally, we examine whether these changes in bank lending translated into real effects at the firm level and assess their implications for aggregate productivity (Section 5).

3. Bank Level Effects

Using the exposure measure described above, we first examine how the liberalization affected bank funding and lending. We estimate the following difference-in-differences specification:

$$Z_{b,t} = \beta_1 Exposure_b \times Post_t + X_b'\theta \times Post_t + \alpha_b + \gamma_t + \epsilon_{b,t}, \tag{1}$$

where $Z_{b,t}$ denotes an outcome for bank b in year t. $Exposure_b$ is the continuous measure of bank exposure to liberalization and $Post_t$ is an indicator equal to one in the post-liberalization period. X_b is a vector of ex-ante bank characteristics, such as bank size and liquidity, that could influence bank-level outcomes. We use the ex-ante values of these characteristics to avoid endogeneity concerns, as the liberalization episode could itself affect these variables. The specification includes bank fixed effects, α_b, which absorb time-invariant differences across banks, and year fixed effects, γ_t, which absorb shocks common to all banks in a given year. Standard errors are clustered at the bank level and p-values are obtained using wild cluster bootstrap inference.

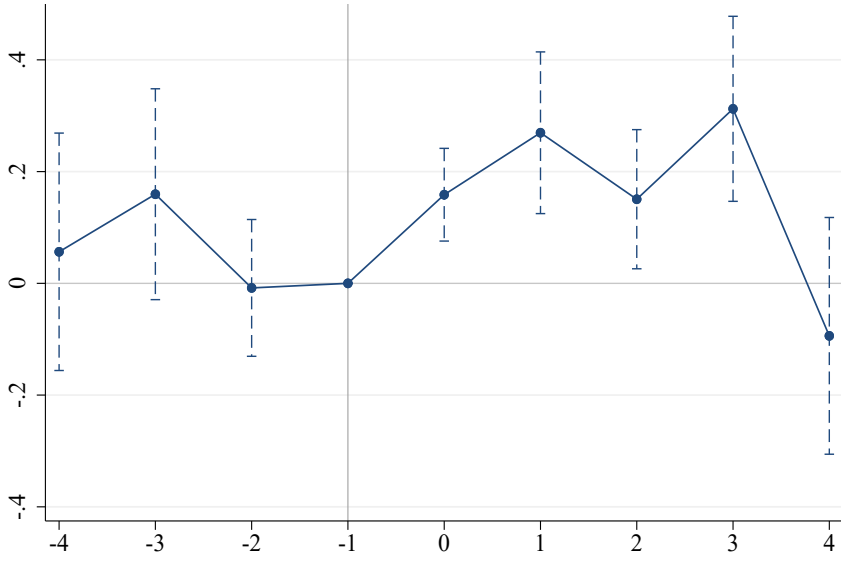
We first ask whether greater exposure to the liberalization expanded banks' access to funding. Table 1 (Panel A) shows that more-exposed banks experienced a substantial increase in borrowing following the reform. A 1 percentage point increase in exposure is associated with an approximately 4.7% increase in total bank borrowing. Importantly, the composition of borrowing also shifted toward foreign sources: the share of borrowing from abroad increased by 0.8 percentage points. Consistent with this shift, foreign borrowing itself increased significantly among more-exposed banks — a 1 percentage point increase in exposure is associated with a 5.7% increase in foreign borrowing. More-exposed banks also raised significantly more capital through total equity issuance, which increased by 7.3% for a 1 percentage point increase in exposure. Taken together, these results show that the liberalization was associated with greater access to external funding through both equity and debt markets.

TABLE 1. Impact of Liberalization on Overall Bank Funding and Lending

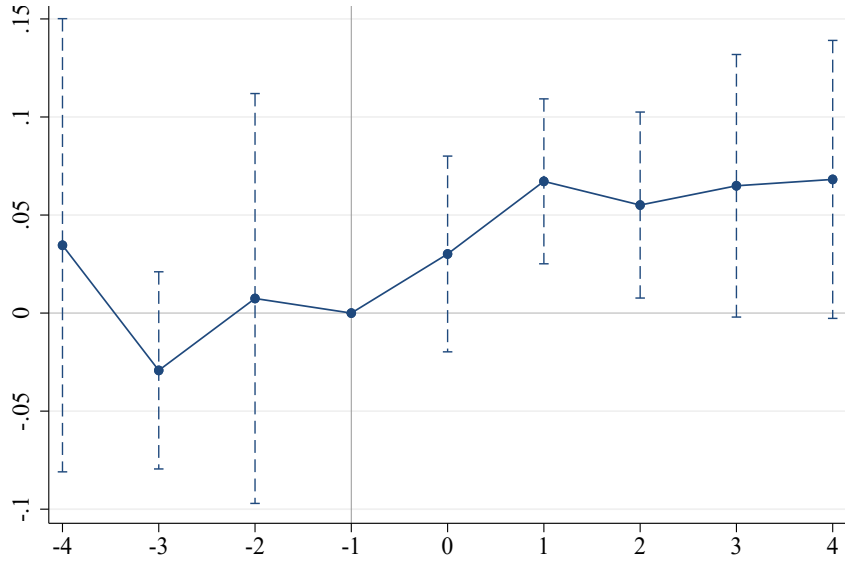
	Total Borrowing	Foreign Borrowing	Foreign Borrowing Share	Total Equity Issuance
Panel A: Bank Funding				
Exposure × Post	0.047*	0.057***	0.789**	0.073*
	(0.017)	(0.013)	(0.247)	(0.029)
Bank Chars. × Post	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	341	255	288	341
Adj. R-squared	0.727	0.756	0.587	0.109
Panel B: Bank Lending				
	Total Advances		Liquidity Ratio	
Exposure × Post	0.041**		0.622**	
	(0.009)		(0.233)	
Bank Chars. × Post	Yes		Yes	
Bank FE	Yes		Yes	
Time FE	Yes		Yes	
Observations	341		341	
Adj. R-squared	0.947		0.684	

Notes: This table reports coefficients of equation 1. Panel A reports the impact of liberalization on Bank Funding: Total Borrowing, Foreign Borrowing, the share of borrowing sourced from outside India and Total Equity Issuance. Panel B reports the impact on Bank Lending: Total Advances and Liquidity (measured using the Loan-Deposit Ratio). All variables are in logarithms except Foreign Borrowing Share and Liquidity Ratio, which are expressed as percentages. All specifications include bank and year fixed effects. Standard errors in parentheses are clustered at the bank level and p-values are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4 examines the dynamics of the two direct measures of external funding — total equity issuance and foreign borrowing. The pre-liberalization coefficients show no evidence of differential pre-trends, while both outcomes increase for more-exposed banks following the liberalization. The dynamic estimates therefore complement the average post-reform effects in Table 1 and show how access to these funding sources evolved around the liberalization.



(A) Bank Equity Issuance



(B) Bank Foreign Borrowings

FIGURE 4. Event Study Graph for the Effect of Liberalization on Bank Funding

Note: This figure reports event study estimates of the effect of liberalization on banks' total equity issuance (Panel A) and foreign borrowing (Panel B), both measured in logarithms. Each point is the coefficient on the interaction between bank exposure and year relative to liberalization, normalized to zero in the year immediately preceding the reform. The confidence interval is at the 95% level.

We next examine whether the expansion in funding was accompanied by an increase in bank lending (Table 1, (Panel B)). On the asset side, a 1 percentage point increase in exposure is associated with an approximately 4.1% increase in total lending following the reform. The loan-deposit ratio also rises by 0.6 percentage points, indicating that lending expanded relative to banks' deposit base. This pattern is consistent with the increase in non-deposit sources of funding documented above allowing more-exposed banks to expand their balance sheets and extend more credit.

Taken together, the bank-level results show that greater exposure to the liberalization expanded banks' access to funding and was accompanied by a substantial increase in aggregate lending. Bank-level lending, however, does not by itself identify an increase in credit supply: differences in aggregate lending across banks could also reflect changes in the composition of their borrowers or differences in credit demand. We therefore turn next to the loan-level data, which allows us to compare lending by differently exposed banks while accounting for borrower-level credit demand by isolating the credit-supply response to the liberalization.

4. Credit Supply Effects

4.1 Intensive Margin

4.1.1 Overall Credit Supply Effects

We begin by examining the intensive margin of credit supply (i.e., when the bank has a pre-existing relationship with a firm) using loan-level data between banks and firms. Following the within-firm estimation strategy of Khwaja and Mian (2008), we focus on firms borrowing from multiple banks and ask whether the same firm receives more credit from banks that are more exposed to the liberalization. By comparing lending across banks to the same firm in a given year, this approach allows us to control for firm-level credit demand. Specifically, we estimate the following difference-in-differences specification:

$$\ln(1+Outstanding\,Credit)_{f,b,t} = \beta_1 Exposure_b \times Post_t + X_b'\theta \times Post_t + \alpha_{f,t} + \gamma_{f,b} + \epsilon_{f,b,t}. \quad (2)$$

The dependent variable is the log of outstanding credit between bank b and firm f in year t. $Exposure_b$ is the continuous measure of bank exposure to equity liberalization defined above and $Post_t$ is an indicator equal to one in the post-liberalization period. As before, we control for ex-ante bank characteristics, including bank size and liquidity, interacted with the post-reform indicator. Firm-year fixed effects, $\alpha_{f,t}$, absorb firm-specific credit demand in each year, so identification comes from comparing the credit supplied to the same firm by banks with different levels of exposure. We also include firm-bank fixed effects, $\gamma_{f,b}$, to absorb persistent features of bank-firm relationships, such as relationship

lending. Standard errors are clustered at the bank level and p-values are obtained using wild cluster bootstrap inference.

The firm-year fixed effects specification restricts the sample to firms with multiple banking relationships. To include firms borrowing from a single bank, we also follow Degryse et al. (2019) and replace firm-year fixed effects with region-sector-year fixed effects. The identifying assumption in this specification is that firms operating in the same region and sector in a given year face similar credit demand shocks. This allows us to use the full sample of firms and provides a complementary way of controlling for credit demand.

Table 2 (Panel A) reports the results. In both specifications, banks more exposed to the liberalization increase their credit supply following the reform. A 1 percentage point increase in bank exposure raises outstanding credit by approximately 1.1% in the firm-year fixed effects specification and by approximately 1.6% in the region-sector-year fixed effects specification. The results are therefore similar across the two approaches, despite the latter including a substantially larger sample of firms. This suggests that the increase in credit supply is not driven solely by firms with multiple banking relationships.

Figure 5A presents the corresponding dynamic estimates. The pre-liberalization coefficients are statistically indistinguishable from zero, providing no evidence of differential pre-trends. Following the reform, credit supply increases and remains persistently higher. Together, the static and dynamic estimates indicate that banks with greater exposure to the liberalization expanded credit supply after the reform.

4.1.2 Firm Productivity and Credit Supply

Having established an increase in credit supply, we next ask how this additional credit was allocated across firms. In particular, we examine whether banks directed relatively more credit toward more productive firms. Following Cingano and Hassan (2022), we measure firm productivity using the marginal revenue product of capital (MRPK).[11] Since collateral is also likely to shape banks' lending decisions, we subsequently examine credit allocation by productivity and collateral jointly, using fixed assets (FA) as a measure of firms' ability to pledge collateral.[12]

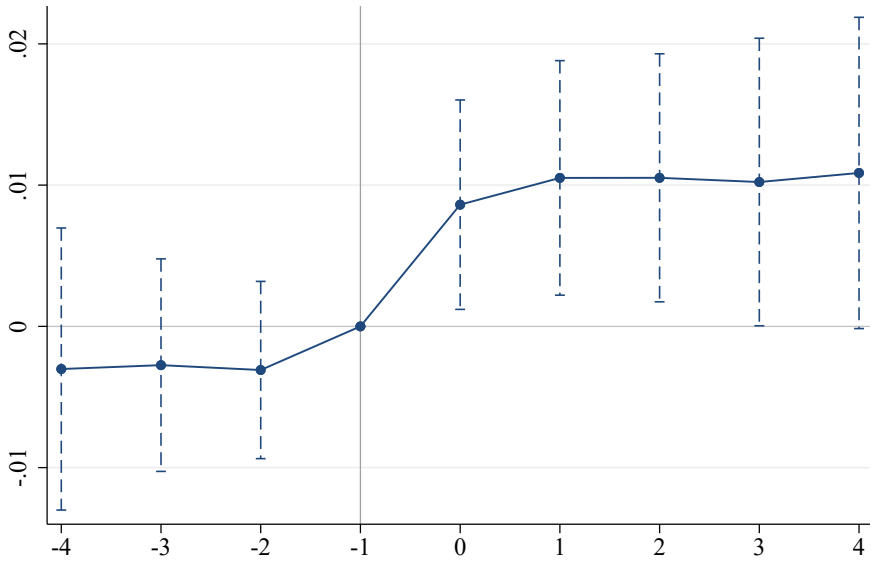
We estimate the following difference-in-differences specification:

$$\ln(1+Outstanding\,Credit)_{f,b,t} = \sum_{d} \beta_d D_f^d (Exposure_b \times Post_t) + X_b'\theta \times Post_t + \alpha_{f,t} + \gamma_{f,b} + \epsilon_{f,b,t} \quad (3)$$

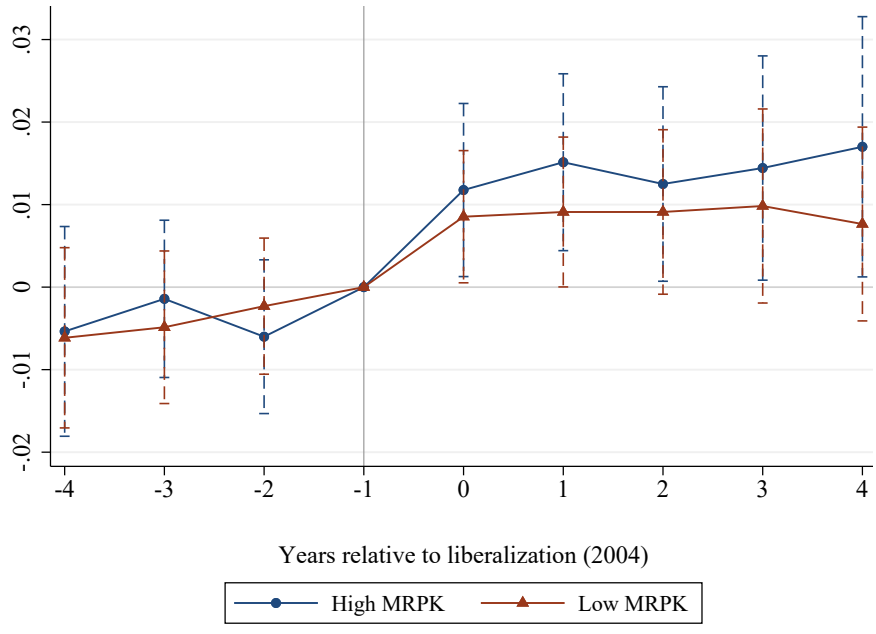
where D_f^d denotes the relevant firm category. We first distinguish between high- and

[11] We measure MRPK as the ratio of sales over gross fixed assets for each firm. See Appendix A for more details.

[12] Firms are classified as high or low MRPK and high or low FA using within-industry medians of their ex-ante values. We refer to firms with ex-ante high-MRPK as more productive firms.



(A) Credit Supply to All Firms



(B) Credit Supply by Firm Productivity

FIGURE 5. Event Study Graph for the Effect of Liberalization on Bank Credit Supply

Note: This figure reports event study estimates of the effect of liberalization on bank credit supply. The dependent variable is log of outstanding credit. Each point is the coefficient on the interaction between bank exposure and year relative to liberalization, normalized to zero in the year immediately preceding the reform. Panel A reports estimates of the effect on credit supply to the average firm. Panel B separates the effects for firms with ex-ante high- and low- MRPK. The confidence interval is at the 95% level.

TABLE 2. Impact of Liberalization on Bank Credit Supply to Firms

Dependent variable: $\ln(1 + \text{Outstanding Credit})$		
	(1)	(2)
Panel A: Average Effect		
Exposure × Post	0.011**	0.016*
	(0.004)	(0.007)
Observations	91,938	145,222
Adj. R-squared	0.721	0.682
Panel B: Heterogeneity by Productivity (MRPK)		
Exposure × Post × High MRPK	0.017**	0.022**
	(0.007)	(0.009)
Exposure × Post × Low MRPK	0.010*	0.016
	(0.006)	(0.009)
High vs Low Differential (t-test)	0.006**	0.006**
	(0.002)	(0.003)
Observations	56,323	65,425
Adj. R-squared	0.710	0.687
Panel C: Heterogeneity by Productivity and Collateral (MRPK × FA)		
Exposure × Post × High MRPK-High FA	0.020***	0.026***
	(0.007)	(0.009)
Exposure × Post × High MRPK-Low FA	0.001	0.010
	(0.009)	(0.010)
Exposure × Post × Low MRPK-High FA	0.011*	0.020**
	(0.006)	(0.009)
Exposure × Post × Low MRPK-Low FA	-0.002	-0.017
	(0.006)	(0.011)
HH vs HL Differential (t-test)	0.019***	0.016**
	(0.008)	(0.008)
Observations	56,297	65,394
Adj. R-squared	0.710	0.688
Bank Chars. × Post	Yes	Yes
Firm-Time FE	Yes	No
Region-Sector-Time FE	No	Yes
Firm-Bank FE	Yes	Yes

Notes: The table reports coefficients of equations 2 and 3. Panel A shows the overall impact of bank exposure to liberalization on credit supply to firms along the intensive margin. Panel B compares firms with high versus low productivity (measured by the ex-ante marginal revenue product of capital, MRPK). Panel C decomposes further by collateral (measured by ex-ante fixed assets, FA); the reported HH vs HL differential is the collateral premium among high-MRPK firms. Column (1) uses firm-time fixed effects and column (2) uses region-sector (NIC 2-digit)-time fixed effects to control for credit demand. Standard errors in parentheses are clustered at the bank level; p-values and significance stars are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

low-MRPK firms and then divide firms into four groups according to productivity and collateral: high MRPK–high FA, high MRPK–low FA, low MRPK–high FA and low MRPK–low FA. Comparing the coefficients across groups allows us to examine whether the expansion in credit following the liberalization was allocated differently across firm types.

Table 2 (Panel B) reports the heterogeneous effects. Panel B shows that credit supply increases more strongly for high-MRPK firms. In the firm-year fixed effects specification, a 1 percentage point increase in exposure raises credit supply by approximately 1.7% for high-MRPK firms compared with 1% for low-MRPK firms. The corresponding estimates using region-sector-year fixed effects are 2.2% and 1.6%, respectively. The high-low differential is positive and statistically significant in both specifications, indicating that more-exposed banks direct relatively more of the additional credit toward more productive firms.

Panel C shows that collateral also matters for the allocation of credit. For high-MRPK firms, the increase in credit supply is concentrated among firms with greater fixed assets. A 1 percentage point increase in exposure raises credit supply by approximately 2% for high-MRPK, high-FA firms in the firm-year fixed effects specification, compared with a small and statistically insignificant increase for high-MRPK, low-FA firms. The same pattern is present when using region-sector-year fixed effects. These results suggest that while banks allocate relatively more of the additional credit toward productive firms, collateral remains an important determinant of which firms receive credit.

Figure 5B presents the dynamic estimates by firm productivity. There is no evidence of differential pre-trends for either high- or low-MRPK firms. Following liberalization, credit supply to high-MRPK firms increases more sharply and persistently than credit supply to low-MRPK firms.

Overall, the intensive-margin results show that the liberalization increased bank credit supply and changed how that credit was allocated across firms. More-exposed banks increased lending relatively more toward firms with higher ex-ante productivity, with collateral further influencing how the additional credit was allocated within productivity groups.

4.2 Extensive Margin

We next examine the extensive margin of credit supply by studying whether banks change the set of firms with which they maintain lending relationships. We consider two outcomes: the likelihood that a bank ends an existing relationship with a firm (Exit) and the likelihood that it initiates a new lending relationship (Entry).

4.2.1 Exit Likelihood

We first examine whether exposure to the liberalization affected banks' decision to maintain existing lending relationships. Among firm-bank relationships that existed prior to the liberalization, we define an exit as the absence of any new loan originations between the bank and the firm in the post-liberalization period.[13] We estimate the following cross-sectional specification at the firm-bank level:

$$Exit_{f,b} = \beta_1 Exposure_b + X_b'\theta + \alpha_f + \epsilon_{f,b}. \tag{4}$$

The dependent variable, $Exit_{f,b}$, is an indicator variable for exit as defined above. $Exposure_b$ is the bank's exposure measure and X_b includes ex-ante bank characteristics such as bank size and liquidity. Firm fixed effects, α_f, allow us to compare the likelihood of exit across banks lending to the same firm. As in the intensive-margin analysis, we also estimate a specification replacing firm fixed effects with region-sector fixed effects, which allows us to include a broader set of firm-bank relationships. Standard errors are clustered at the bank level and p-values are obtained using wild cluster bootstrap inference.

Table 3 (Panel A) reports the results. On average, more exposed banks are less likely to end existing lending relationships following the liberalization, although the effect is not statistically significant in either specification. We next examine whether this average effect masks differences across firm types.

To do so, we allow the effect of bank exposure to vary across the firm categories D_f^d as defined earlier:

$$Exit_{f,b} = \beta_1 \sum_d \beta_d D_f^d Exposure_b + X_b'\theta + \alpha_f + \epsilon_{f,b}. \tag{5}$$

The heterogeneous effects are significant mainly in the region-sector fixed effects specification, so we focus on these estimates. In this specification, more-exposed banks are significantly less likely to end relationships with high-MRPK firms than with low-MRPK firms. A 1 percentage point increase in bank exposure is associated with a 0.5 percentage point decline in the probability of exit for high-MRPK firms, compared with an insignificant 0.2 percentage point decline for low-MRPK firms. The difference between the two groups is statistically significant.

The joint productivity-collateral results reveal further differences in exit across firm types. Firms with both low MRPK and low FA are significantly more likely to experience an exit when borrowing from more-exposed banks. Among high-MRPK firms, more-exposed banks are also less likely to end relationships with high-FA firms than with low-FA firms. Overall, while the average exit effect is not statistically significant, the heterogeneous results

[13] The bank and firm may still have an ongoing relationship in the sense that loans originated before the liberalization remain outstanding.

TABLE 3. Impact of Liberalization on Likelihood of Exit

Dependent variable: Exit Dummy		
	(1)	(2)
Panel A: Average Effect		
Bank Exposure	-0.0022	-0.0028
	(0.0012)	(0.0017)
Observations	8,927	13,615
Adj. R-squared	0.2751	0.0665
Panel B: Heterogeneity by Productivity (MRPK)		
Exposure × High MRPK	-0.0019	-0.0049**
	(0.0014)	(0.0020)
Exposure × Low MRPK	-0.0023	-0.0015
	(0.0014)	(0.0018)
High vs Low Differential (t-test)	0.0005	-0.0033***
	(0.0006)	(0.0014)
Observations	5,968	7,033
Adj. R-squared	0.2788	0.0873
Panel C: Heterogeneity by Productivity and Collateral (MRPK × FA)		
Exposure × High MRPK-High FA	-0.0019	-0.0051**
	(0.0015)	(0.0018)
Exposure × High MRPK-Low FA	-0.0016	-0.0034
	(0.0019)	(0.0023)
Exposure × Low MRPK-High FA	-0.0028	-0.0032
	(0.0014)	(0.0017)
Exposure × Low MRPK-Low FA	0.0035**	0.0083***
	(0.0030)	(0.0021)
HH vs HL Differential (t-test)	-0.0003	-0.0017*
	(0.0020)	(0.0016)
Observations	5,964	7,029
Adj. R-squared	0.2788	0.0941
Bank Chars.	Yes	Yes
Firm FE	Yes	No
Region-Sector FE	No	Yes

Notes: This table reports coefficients from equations 4 and 5 which estimate cross-sectional regressions of the exit dummy on bank exposure (Panel A) and bank exposure interacted with firm type (Panels B–C). The sample is a single cross-section of firm-bank pairs with a relationship established before 2004; the dependent variable equals one if no new loan is originated after 2004. Panel B compares firms with high versus low productivity (measured by the ex-ante marginal revenue product of capital, MRPK). Panel C decomposes further by collateral (measured by ex-ante fixed assets, FA); the reported HH vs HL differential is the collateral premium among high-MRPK firms. Column (1) uses firm fixed effects and column (2) uses region-sector (NIC 2-digit) fixed effects to control for credit demand. Standard errors in parentheses are clustered at the bank level; p-values and significance stars are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

suggest that exit decisions by more-exposed banks varied according to firm productivity and collateral.

4.2.2 Entry Likelihood

We next examine whether exposure to the liberalization affected banks' decisions to form new lending relationships. We construct a two-period panel of all potential firm-bank pairs in the sample, covering the pre- and post-liberalization periods.[14] We estimate the following two-period difference-in-differences specification:

$$Entry_{f,b,\tau} = \beta_1 Exposure_b \times Post_\tau + X_b'\theta \times Post_\tau + \alpha_{f,\tau} + \gamma_b + \epsilon_{f,b,\tau}. \tag{6}$$

The dependent variable, $Entry_{f,b,\tau}$, indicates whether bank b initiates a new lending relationship with firm f in period τ. $Exposure_b$ is the bank's exposure measure and X_b includes ex-ante bank characteristics such as bank size and liquidity. The specification includes firm-period fixed effects and bank fixed effects. As in the intensive-margin analysis, we also estimate a specification replacing firm-period fixed effects with region-sector-period fixed effects. Standard errors are clustered at the bank level and p-values are obtained using wild cluster bootstrap inference.

Table 4 (Panel A) reports the results. More-exposed banks are more likely to form new lending relationships after the liberalization. A 1 percentage point increase in bank exposure raises the probability of entry by approximately 0.05 percentage points in both specifications.

We next examine whether this effect differs across firms by allowing the effect of bank exposure to vary with the firm categories D_f^d defined above:

$$Entry_{f,b,\tau} = \sum_d \beta_d D_f^d \left(Exposure_b \times Post_\tau\right) + X_b'\theta \times Post_\tau + \alpha_{f,\tau} + \gamma_b + \epsilon_{f,b,\tau}. \tag{7}$$

The results show that the increase in entry is concentrated among high-MRPK firms. A 1 percentage point increase in bank exposure raises the probability of forming a new relationship with a high-MRPK firm by approximately 0.08 percentage points in both specifications. In contrast, the effect for low-MRPK firms is close to zero. The difference between high- and low-MRPK firms is statistically significant.

The joint productivity-collateral results show a similar pattern. The increase in entry is largest for firms with both high MRPK and high FA: a 1 percentage point increase in exposure raises their probability of entry by approximately 0.1 percentage point. By contrast, more-exposed banks are significantly less likely to form new relationships with firms that

[14] Pairs that never form a lending relationship are coded zero in both periods, while pairs that enter before or after the liberalization are coded one in the corresponding period.

TABLE 4. Impact of Liberalization on Likelihood of Entry

Dependent variable: Entry Dummy		
	(1)	(2)
Panel A: Average Effect		
Bank Exposure × Post	0.0005**	0.0005**
	(0.0002)	(0.0002)
Observations	1,485,088	1,264,501
Adj. R-squared	0.0422	0.0170
Panel B: Heterogeneity by Productivity (MRPK)		
Exposure × Post × High MRPK	0.0008*	0.0008***
	(0.0003)	(0.0003)
Exposure × Post × Low MRPK	0.0000	-0.0000**
	(0.0002)	(0.0002)
High vs Low Differential (t-test)	0.0008***	0.0008***
	(0.0001)	(0.0001)
Observations	400,383	322,684
Adj. R-squared	0.0589	0.0279
Panel C: Heterogeneity by Productivity and Collateral (MRPK × FA)		
Exposure × Post × High MRPK-High FA	0.0010*	0.0011***
	(0.0003)	(0.0003)
Exposure × Post × High MRPK-Low FA	0.0006	0.0005
	(0.0002)	(0.0002)
Exposure × Post × Low MRPK-High FA	0.0002	0.0004
	(0.0003)	(0.0003)
Exposure × Post × Low MRPK-Low FA	-0.0004***	-0.0006***
	(0.0002)	(0.0002)
HH vs HL Differential (t-test)	0.0005*	0.0007***
	(0.0002)	(0.0002)
Observations	400,017	322,439
Adj. R-squared	0.0591	0.0284
Bank Chars. × Post	Yes	Yes
Firm-Time FE	Yes	No
Region-Sector-Time FE	No	Yes
Bank FE	Yes	Yes

Notes: The table reports the coefficients of equations 6 and 7. It shows the impact of bank exposure to liberalization on the likelihood of a bank forming a new relationship with a firm (Panel A) and according to firm type (Panels B–C). The dependent variable Entry is defined with a pre-period of 1998–2003 (Period 1) and a post-period of 2004–2008 (Period 2). It equals one in the period a firm-bank pair first forms a lending relationship and zero in periods it has not yet formed one; pairs that already had a relationship as of period 1 are excluded from period 2. Column 1 uses firm-time fixed effects and Column 2 uses region-sector (NIC 2-digit)-time fixed effects to control for credit demand; both columns also include bank fixed effects. Panel B compares firms with high versus low productivity (measured by the ex-ante marginal revenue product of capital (MRPK)). Panel C decomposes jointly by MRPK and FA. The reported differential (HH vs HL) is the collateral premium among high-MRPK firms. Standard errors in parentheses are clustered at the bank level and p-values are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

have both low MRPK and low fixed assets. Among high-MRPK firms, the likelihood of entry also increases more for high-FA firms than for low-FA firms. Overall, the entry results show that more-exposed banks formed relatively more new relationships with productive and well-collateralized firms.

We also examine how much credit more-exposed banks extend once a new relationship is formed. These results are reported in Appendix C. We find that more-exposed banks extend smaller initial loans to new borrowers, with some evidence that lending increases as these relationships mature. This pattern is consistent with banks screening new borrowers cautiously at the start of a relationship and expanding credit as they learn more about borrower quality.

In sum, along both intensive and extensive margins, more-exposed banks increased credit allocation to more productive firms than to less productive firms (compared to less-exposed banks).

5. Real Effects

5.1 Firm Level Effects

Having shown that the liberalization increased credit supply and changed how credit was allocated across firms, we next examine whether these changes affected firm-level outcomes. We first construct a measure of each firm's exposure to the banking-sector liberalization. We define firm exposure as the weighted average of the exposure of the banks from which the firm borrowed at the time of the reform [15]. The weights are given by each bank's share of the firm's outstanding credit:

$$Firm\ Exposure_f = \sum_b Exposure_b \frac{Outstanding\ Credit_{f,b}}{Total\ Outstanding\ Credit_f}. \tag{8}$$

A firm is therefore more exposed to the liberalization if a larger share of its credit comes from more-exposed banks. A potential concern is that firms with higher exposure might be systematically different from less exposed firms, and these differences, rather than the reform could be driving the firm-level results. Figure B1B plots standardized coefficients from regressions of firm exposure on pre-reform firm characteristics. While exposure is related to firm size and the wage-to-assets ratio, it is not significantly related to bank borrowing, sales-to-assets or pre-reform MRPK. We account for this issue by using firm fixed effects in our specifications which absorb persistent time invarying differences across firms. Further, the dynamic specifications below provide an additional check for differential

[15] Because firm exposure is constructed using lending relationships in place at the time of the liberalization, the firm-level analysis is restricted to firms with an existing bank relationship prior to the reform.

pre-reform trends. [16]

We use this measure to estimate the following firm-level difference-in-differences specification:

$$\ln(Z_{f,t}) = \beta_1 \textit{Firm Exposure}_f \times Post_t + X_f'\theta \times Post_t + \alpha_{r,s,t} + \gamma_f + \epsilon_{f,t}. \quad (9)$$

The outcomes $Z_{f,t}$ include bank borrowing, fixed assets, the wage bill and MRPK. Fixed assets capture firm investment, while the wage bill captures employment. Bank borrowing is measured using firm balance sheet data and therefore provides an additional check on the increase in credit found in the loan-level analysis. X_f is a weighted average of ex-ante characteristics of the firm's banks, constructed using the same credit weights as the firm exposure measure. The specification includes firm fixed effects and region-sector-year fixed effects. Standard errors are clustered at the two-digit industry level and p-values are obtained using wild cluster bootstrap inference.

Table 5 (Panel A) reports the average effects. Firms with greater exposure to the liberalization experience a significant increase in bank borrowing after the reform. A 1 percentage point increase in firm exposure raises bank borrowing by approximately 1 percent. This is consistent with the increase in credit documented in the loan-level analysis. For the average firm, the effects on fixed assets, the wage bill and MRPK are positive but not statistically significant.

The average effects, however, mask important differences between firms with high and low productivity (MRPK). To examine this heterogeneity, we allow the effect of firm exposure to vary across the firm categories defined earlier:

$$\ln(Z_{f,t}) = \sum_d \beta_d D_f^d \left(\textit{Firm Exposure}_f \times Post_t \right) + X_f'\theta \times Post_t + \alpha_{r,s,t} + \gamma_f + \epsilon_{f,t}. \quad (10)$$

Table 5 (Panels B and C) reports the results. The increase in bank borrowing is concentrated among high-MRPK firms. These firms experience a significant increase in borrowing after the liberalization, while the effect for low-MRPK firms is small and not statistically significant. The real effects show a similar pattern. High-MRPK firms increase both fixed assets and their wage bill, indicating an expansion in investment and employment. Low-MRPK firms do not experience a similar expansion and, in some cases, contract. These results suggest that the additional credit following the liberalization allowed firms with higher marginal returns to capital to expand.

The joint productivity-collateral results further show that these effects are not limited to high-MRPK firms with large amounts of collateral. High-MRPK firms with relatively low FA also experience increases in borrowing and real activity. The effects on investment and

[16] The lack of a relationship between firm exposure and pre-reform MRPK is also reassuring for the key results on misallocation.

TABLE 5. Impact of Liberalization on Firm Outcomes

	Dependent variable			
	(1) Bank Borrowing	(2) Fixed Assets	(3) Wage Bill	(4) MRPK
Panel A: Average Effect				
Exposure × Post	0.010**	0.001	0.002	0.001
	(0.004)	(0.003)	(0.002)	(0.003)
Observations	34,500	40,438	38,392	37,676
Adj. R-squared	0.790	0.918	0.903	0.778
Panel B: MRPK				
Exposure × Post × High MRPK	0.020***	0.010***	0.006**	-0.008***
	(0.004)	(0.003)	(0.002)	(0.003)
Exposure × Post × Low MRPK	-0.001	-0.011**	-0.003	0.012***
	(0.004)	(0.003)	(0.003)	(0.004)
High vs Low Differential (t-test)	0.021***	0.020***	0.009**	-0.020***
	(0.004)	(0.002)	(0.003)	(0.004)
Observations	29,494	33,988	32,803	32,463
Adj. R-squared	0.784	0.917	0.899	0.766
Panel C: Joint MRPK and FA				
Exposure × Post × High MRPK-High FA	0.015**	0.004	0.002	-0.005*
	(0.005)	(0.004)	(0.002)	(0.003)
Exposure × Post × High MRPK-Low FA	0.028***	0.017***	0.011***	-0.010*
	(0.004)	(0.003)	(0.003)	(0.005)
Exposure × Post × Low MRPK-High FA	-0.001	-0.010**	-0.003	0.010**
	(0.004)	(0.003)	(0.003)	(0.004)
Exposure × Post × Low MRPK-Low FA	0.002	-0.010**	-0.003	0.020**
	(0.009)	(0.005)	(0.005)	(0.009)
HH vs HL Differential (t-test)	-0.012***	-0.012***	-0.009	0.005***
	(0.005)	(0.003)	(0.003)	(0.005)
Observations	29,471	33,965	32,777	32,439
Adj. R-squared	0.784	0.917	0.899	0.766
Bank Chars. × Post	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Region-Sector-Time FE	Yes	Yes	Yes	Yes

Note: This table reports the coefficients of equations 9 and 10. It shows how liberalization impacted firm-level credit (secured bank borrowings), firm investment (proxied by fixed assets), firm employment (proxied by wage bill), and firm MRPK. Panel A reports the overall effect; Panels B and C report effects by firm type. Panel B compares firms with high versus low productivity (measured by the ex-ante marginal revenue product of capital, MRPK). Panel C decomposes further by collateral (measured by ex-ante fixed assets, FA). The reported differential in Panel C (HH vs HL) is the collateral premium among high-MRPK firms. All variables are expressed in logarithms. Standard errors in parentheses are clustered at the industry (NIC 2-digit) level and p-values are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

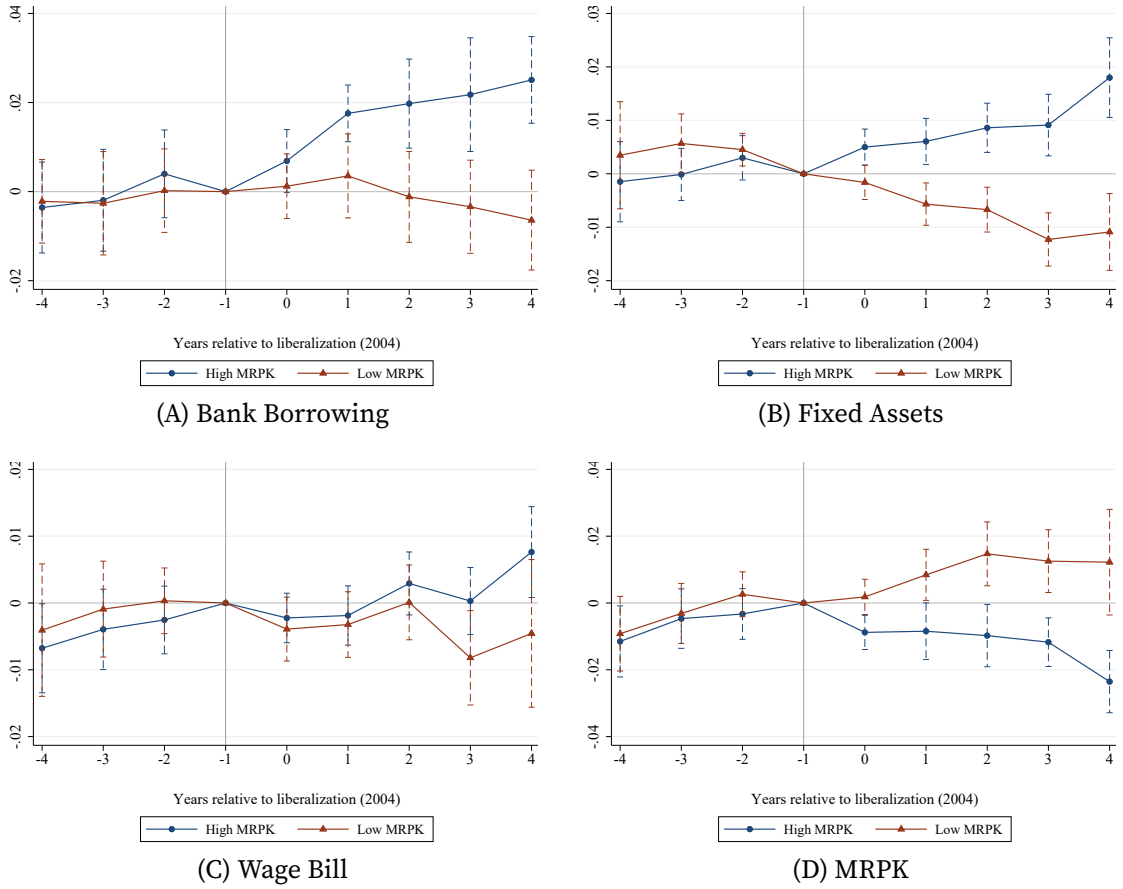


FIGURE 6. Event Study Graphs for the Effect of Liberalization on Firm Outcomes

Note: This figure reports event study estimates of the effect of liberalization on various firm outcomes: bank borrowing (Panel A), fixed assets (Panel B), the wage bill (Panel C) and MRPK (Panel D), separately for ex-ante high- and low-MRPK firms. All dependent variables are in logarithms. Each point is the coefficient on the interaction between firm exposure and year relative to liberalization, normalized to zero in the year immediately preceding the reform. The confidence intervals are at the 95% level.

employment are particularly large for this group, consistent with the liberalization relaxing credit constraints for firms with high returns to capital but fewer assets to pledge.

Impact on Firm Productivity. We next examine how these changes are reflected in the evolution of MRPK. A 1 percentage point increase in firm exposure lowers MRPK by approximately 0.8 percent among initially high-MRPK firms and raises it by approximately 1.2 percent among initially low-MRPK firms. The gap in MRPK between the two groups therefore narrows after the liberalization. This pattern is consistent with capital moving toward firms with initially high marginal returns and away from firms with initially low marginal returns. The resulting convergence in MRPK points to a decline in capital misallocation following the liberalization.

Figure 6 presents the dynamic estimates separately for high- and low-MRPK firms. The

pre-reform coefficients are small and statistically insignificant, thus indicating that there were no differential pre-trends. After the reform, bank borrowing and fixed assets rise more for high-MRPK firms than for low-MRPK firms. The wage bill shows a similar, though more modest, divergence. MRPK moves in the opposite direction across the two groups: it falls for initially high-MRPK firms and rises for initially low-MRPK firms. The dynamic effect results reinforce the evidence that the liberalization reduced the dispersion in marginal returns to capital and improved the allocation of capital across firms.

5.2 Aggregate Effects

The firm-level results show that the liberalization shifted credit towards firms with higher marginal returns to capital and reduced the dispersion of MRPK. These patterns point to an improvement in the allocation of resources across firms. We now quantify the aggregate productivity gains associated with this reallocation.

Following the literature, we model misallocation as firm-specific wedges on input prices. For firm i, the effective price of input x is $(1+\tau_i^x)p^x$, where $x \in (K, L, M)$ denotes capital, labor, and materials, respectively; p^x is the observed price of the input; and τ_i^x is the firm-specific wedge.[17] These wedges generate differences in marginal products across firms and therefore create scope for aggregate productivity gains through reallocation.

Our objective is to isolate this reallocation channel from changes in firms' underlying productivity. Intuitively, even if firm-level productivity remains unchanged, aggregate output can increase if the liberalization causes inputs to move toward firms where their marginal products are higher. To quantify these gains, we follow the aggregation framework of Bau and Matray (2023), which maps reform-induced changes in firms' input use, together with their pre-existing distortions, into changes in aggregate productivity.[18] We use the change in the aggregate Solow residual as our measure of the change in aggregate productivity.

We decompose the change in the aggregate Solow residual into two components — changes in firms' own productivity and gains from the reallocation of inputs across firms:

$$\Delta Solow_{I,t} \approx \sum_{i \in I} \lambda_i \Delta \log A_i + \sum_{\substack{i \in I \\ x \in \{K,L,M\}}} \lambda_i \alpha_i^x \frac{\tau_i^x}{1+\tau_i^x} \Delta \log x_i. \tag{11}$$

Here, λ_i captures the importance of firm i in aggregate output, $\Delta \log A_i$ is the change in firm-level productivity, α_i^x is the output elasticity of input x, τ_i^x is the firm-specific wedge on that input [19] and $\Delta \log x_i$ is the change in the firm's use of the input. The first term captures

[17] The wedge τ_i^x can be positive or negative. A positive wedge can be interpreted as a tax on the input relative to its observed price, while a negative wedge can be interpreted as a subsidy.

[18] Bau and Matray (2023) develop this approach in the context of India's staggered FDI liberalization across manufacturing industries.

[19] The wedge measures the deviation of the marginal revenue product of input x (capital, labor) from its price

changes in firms' own productivity, while the second captures the effect of reallocating inputs across firms with different wedges.

Because our interest is in the reallocation induced by changes in bank credit, we hold firm-level productivity fixed by setting $\Delta \log A_i = 0$. We also focus on only capital and labor inputs, treating materials as undistorted. The expression relevant for our exercise is therefore:

$$\Delta Solow \approx \sum_{\substack{i \in I \\ x \in \{K,L\}}} \lambda_i \alpha_i^x \frac{\tau_i^x}{1+\tau_i^x} \Delta \log x_i. \tag{12}$$

Unlike the discrete industry-level treatment studied by Bau and Matray (2023), exposure to the banking liberalization varies continuously across firms in our setting. We therefore apply the aggregation exercise to the full sample of firms and construct λ_i as firm i's share of total sample sales. For each firm, the calculation requires three objects: its input elasticities, its pre-reform capital and labor wedges, and the reform-induced changes in its capital and labor inputs. We describe the construction of each below.

We estimate the capital and labor elasticities separately by industry using the Levinsohn and Petrin (2003); Wooldridge (2009) control-function approach, with materials expenditure as the productivity proxy. Firms within an industry therefore share the same estimated input elasticities. [20]

To construct wedges τ_i, we follow the conventional approach in the misallocation literature, as implemented by Bau and Matray (2023). For each firm, we compute the pre-reform MRPK and MRPL [21]. We then use these MRPK and MRPL values to calculate firm-level wedges by calculating $\tau_i^K = \alpha_i^K \frac{p_i y_i}{r K_i} - 1$ and $\tau_i^L = \alpha_i^L \frac{p_i y_i}{w L_i} - 1$ where r is the rental rate of capital and wL_i is the wage bill. We follow Bau and Matray (2023) and Hsieh and Klenow (2009) and set $r = 10\%$.

The final ingredient is the reform-induced change in each firm's capital and labor inputs. We estimate these changes using firm-level difference-in-differences regressions that allow the response to vary with firms' ex-ante MRPK and MRPL:

$$\begin{aligned}\ln(Z_{f,t}) = \beta_1(\textit{Firm Exposure}_f \times Post_t) + \beta_2(HighMRPK_f \times \textit{Firm Exposure}_f \times Post_t) \\ + \beta_3(HighMRPL_f \times \textit{Firm Exposure}_f \times Post_t) + X_f'\theta \times Post_t + \alpha_{r,s,t} + \gamma_f + \epsilon_{f,t}.\end{aligned} \tag{13}$$

Here, $Z_{f,t}$ equals firm level capital and labor inputs measured using gross fixed assets and the wage bill respectively. *Firm Exposure*$_f$ is a firm's exposure to banking liberalization

(rental rate (r), wage (w)) at the initial allocation. If factors are optimally allocated, then the wedge will be zero.

[20] The elasticities are estimated over the full sample period. Industries with insufficient observations for reliable estimation or with estimated elasticities outside the $(0, 1)$ range are assigned the observation-weighted average elasticity from industries with well-behaved estimates.

[21] The pre-reform MRPK and MRPL are 1998-2003 averages and are winsorized at the 1st and 99th percentiles.

as described earlier, $Post_t$ is a dummy for the post-liberalization period, *High MRPK* is an indicator variable for firms classified as ex-ante high-MRPK and *High MRPL* is an indicator variable for firms classified as ex-ante high-MRPL. X_f is a weighted average of bank characteristics. The specification includes firm fixed effects and region-sector-year fixed effects and standard errors are clustered at the two-digit industry level. The results of this regression are presented in Table 6 (Panel A) . We use the coefficients from this regression to calculate predicted change in capital and labor: $\widehat{\Delta \log K_i}$ and $\widehat{\Delta \log L_i}$ for each firm. [22]

TABLE 6. Impact of Liberalization on Aggregate Productivity

Panel A: Impact on Firm Inputs				
	Dependent variable			
	(1) Fixed Assets	(2) MRPK	(3) Wage Bill	(4) MRPL
Exposure × Post	-0.013***	0.015***	-0.013***	0.015***
	(0.003)	(0.005)	(0.003)	(0.003)
Exposure × Post × High MRPK	0.019***	-0.018***	0.004	-0.004
	(0.002)	(0.004)	(0.003)	(0.003)
Exposure × Post × High MRPL	0.005**	-0.006**	0.020***	-0.020***
	(0.003)	(0.003)	(0.003)	(0.003)
Observations	33,758	32,283	32,741	31,981
Adj. R-squared	0.916	0.766	0.899	0.791
Bank Chars. × Post	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Region-Sector-Time FE	Yes	Yes	Yes	Yes

Panel B: Aggregate Productivity Effects	
Channel	Contribution to Δ Solow Residual (%)
Capital (K)	0.74
Labor (L)	0.46
Total	**1.20**

Note: Panel A reports coefficients from equation 13. It shows how liberalization impacts firm-level inputs: capital and labor (proxied by fixed assets and wage bill respectively). It also shows how MRPK and MRPL evolve for firms. All variables are expressed in logarithms. Standard errors in parentheses are clustered at the industry (NIC 2-digit) level. Panel B reports the implied contribution of capital and labor reallocation to aggregate productivity, using the coefficients from Panel A. For each firm, we use its Panel A coefficients, MRPK/MRPL classification and exposure measure to calculate its predicted change in capital and labor. Each firm's contribution to the Solow residual is then computed by weighting these predicted changes by the firm's revenue share and factor wedges; the reported estimate is the sum of these firm-level contributions across the full sample. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Each firm's contribution to the Solow residual is then $\sum_{x \in \{K,L\}} \lambda_i \alpha_i^x \frac{\tau_i^x}{1+\tau_i^x} \Delta log x_i$ and the reported aggregate estimate is the sum of these firm-level contributions across the full

[22] Specifically, we calculate $\widehat{\Delta \log K_i} = \left(\hat{\beta}_1 + \hat{\beta}_2 \cdot HighMRPK_i + \hat{\beta}_3 \cdot HighMRPL_i\right) \times \text{Exposure}_i$ using the firm's MRPK and MRPL classification and its exposure measure and analogously calculate $\widehat{\Delta \log L_i}$ using the wage-bill regression

sample. Table 6 (Panel B) reports these estimates. The results show that exposure to banking sector liberalization increased aggregate productivity for these firms by approximately 1.2% with 0.74 percentage points attributed to improvement in the allocation of capital and 0.46 percentage points to an improvement in labor allocation. To put this in perspective, the aggregate economy wide TFP growth for India in this period was 0.9% and the manufacturing sector TFP growth was 1.6% [23].

These estimates quantify the productivity gains arising specifically from the reallocation of capital and labor following the liberalization. They therefore provide an aggregate counterpart to the firm-level results above: the movement of resources across firms following the reform was large enough to generate economically meaningful productivity gains. [24]

6. Mechanisms

The empirical analysis so far shows that banks not only increased their credit supply following the liberalization but also made better credit allocation decisions. More-exposed banks increased lending disproportionately to more productive firms and this improvement in credit allocation translated into real effects: more productive firms expanded, the dispersion of marginal revenue product of capital across firms declined and aggregate productivity increased. An important question then, is why the expansion in credit following the liberalization was accompanied by an improvement, rather than a deterioration, in the allocation of lending.

The analysis suggests that liberalization led to more than a simple relaxation of banks' funding constraints. More-exposed banks also appear to have changed the way in which they screened, originated, and managed loans. An examination of the volume of lending at the extensive margin in Appendix C provides support for this behavior. When forming new lending relationships, more-exposed banks initially extend smaller loans and subsequently increase their lending as the relationship develops — a pattern consistent with prudent screening and learning about borrower quality. We provide two additional pieces of evidence in this section. First, more-exposed banks increase their investments in human capital following the reform. Second, loans made by more-exposed banks are less likely to be restructured in the future. We then ask why liberalization may have generated these changes in bank behavior and examine whether the form in which capital entered the bank — equity ownership vs. foreign borrowing can help account for the improvement in credit

[23] Source: RBI KLEMS Database

[24] As a complementary aggregation exercise, we also apply the sufficient-statistics approach of Sraer and Thesmar (2023). This method uses reform-induced changes in the distribution of MRPK across sectors to quantify their implications for aggregate TFP. The results, reported in Appendix D, show a significant decline in within-sector MRPK dispersion among more exposed sectors and imply a positive aggregate TFP effect, with most of the gain coming from improved allocation within sectors. This approach measures the aggregate consequences of changes in the MRPK distribution at the sector level. We,therefore, treat it separately from the firm-level input-reallocation exercise above.

allocation decisions.

6.1 Bank Exposure and Investment in Human Capital

Liberalization reduces funding constraints for banks, which may allow them to allocate more time and resources to screening loans and making better-informed lending decisions. Foreign participation may also affect corporate governance and management practices through the transfer of technical and managerial know-how or by increasing the incentives to adopt international banking practices. In addition, greater access to funding may enable banks to invest more in their employees and in the capacity required to screen and monitor borrowers. To explore this further, we estimate the following regression:

$$Z_{b,t} = \beta_1 Exposure_b \times Post_t + X_b'\theta \times Post_t + \alpha_b + \gamma_t + \epsilon_{b,t} \tag{14}$$

Here $Z_{b,t}$ is either the ratio of staff expenses to total income or the log of the number of employees for bank *b* in year *t*. $Exposure_b$ denotes the bank's exposure to equity liberalization, as defined earlier. The specification also controls for ex-ante bank characteristics and bank and year fixed effects. Standard errors are clustered at the bank level and p-values are estimated from a wild cluster bootstrap inference procedure.

The results are presented in Table 7 and show that banks with greater exposure to liberalization spend more on their employees, as reflected in higher staff expenses as well as total number of employees. Specifically, a 1 percentage point increase in bank exposure leads to a 0.12 percentage point increase in staff expenses as a proportion of total income and a 5.5% larger increase in employee headcount. This pattern reflects greater investments in human capital and staff training by more-exposed banks in the post-liberalization period.

TABLE 7. Liberalization and Investment in Human Capital

	Staff Expenses to Total Income	Number of Employees
Exposure × Post	0.122*	0.054**
	(0.053)	(0.012)
Bank Chars. × Post	Yes	Yes
Bank FE	Yes	Yes
Time FE	Yes	Yes
Observations	213	320
Adj. R-squared	0.853	0.924

Notes: The table reports coefficients of equation 14. It shows the impact of bank exposure to liberalization on their human capital investment. Staff Expenses are measured as a percentage of Total Income and Number of Employees is measured in logarithms. Standard errors in parentheses are clustered at the bank level and p-values are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

These results provide one potential link between the relaxation of banks' funding constraints and their improved credit allocation decisions. Greater staff capacity and investment in employees may have increased banks' ability to collect information, evaluate borrowers, and monitor loans. The results do not allow us to distinguish whether the increase in human capital reflects greater resources available for screening, changes in management practices associated with foreign ownership, or both. Nevertheless, they indicate that the post-reform response of more-exposed banks involved changes in bank inputs and internal capacity in addition to an expansion in the volume of lending.

6.2 Bank Exposure and Loan Performance

We next examine whether loans made by more-exposed banks also performed differently following liberalization. The loan-level data record whether a particular loan was restructured or modified. We use this as an indicator of subsequent loan performance and examine whether restructuring rates were different for loans made by more-exposed banks. Using the full sample of loans, we estimate the following difference-in-differences regression:

$$LoanRestructured_{f,b} = \beta_1 Exposure_b \times Post_t + X_b'\theta \times Post_t + \alpha_{f,post} + \epsilon_{f,b} \qquad (15)$$

Here, the dependent variable is a dummy that equals one if the loan between firm *f* and bank *b* was restructured and zero otherwise. $Exposure_b$ denotes the bank's exposure to equity liberalization as defined earlier and $Post_t$ is a dummy for the post-liberalization period. The specification includes firm-period fixed effects. Identification therefore comes from comparing loans made to the same firm by banks with different exposure to the liberalization and asking whether the relative restructuring probability of loans from more exposed banks changed after the reform.

This distinction is important. Because borrower characteristics are held fixed, this regression does not capture the selection of more productive firms documented in the main credit-allocation results. Instead, it asks whether more-exposed banks originated or managed loans differently even conditional on lending to the same borrower. A lower restructuring rate is therefore consistent with differences in underwriting, monitoring, or loan management rather than simply differences in borrower composition.

We also examine the likelihood of loans being restructured for loans that were originated in the post-liberalization period. Specifically, we estimate the following cross-sectional regression:

$$LoanRestructured_{f,b} = \beta_1 Exposure_b + X_b'\theta + \alpha_f + \epsilon_{f,b} \qquad (16)$$

This specification looks only at the loans made in the post-liberalization period. Firm fixed effects again allow us to compare loans made to the same borrower by differently

exposed banks. We estimate this specification both for all post-reform relationships and for relationships that did not exist prior to liberalization.

TABLE 8. Bank Exposure to Liberalization and Loan Restructuring

Dependent variable: Loan Restructured Dummy			
	All Loans	Post- Reform Loans	
	(1) Full Panel	(2) All Relationships	(3) New Relationships
Exposure × Post	-0.0030*	–	–
	(0.0011)		
Exposure	–	-0.0078***	-0.0076***
		(0.0012)	(0.0011)
Bank Chars. × Post	Yes	–	–
Firm-Period FE	Yes	–	–
Firm FE	–	Yes	Yes
Observations	9,806	5,690	3,061
Adj. R-squared	0.1831	0.1667	0.1185

Notes: This table presents the results from equations 15 and 16. The dependent variable is a dummy equal to one if a loan between a firm and a bank is restructured. Column (1) uses the full panel; the coefficient measures whether the probability of a loan being restructured changed in the post-liberalization period for banks more exposed to liberalization, identified from firm-by-period fixed effects (comparing, within the same firm, loans made by banks with differing exposure). Columns (2)–(3) use only loans originated after liberalization, so the coefficient measures the relationship between a bank's exposure and the probability that its post-liberalization loans to a given firm are restructured; both columns absorb firm fixed effects, with identification coming again from comparing, within the same firm, loans made by banks with differing exposure. Column (2) uses all post-liberalization loans; column (3) restricts to firms with no lending relationship in the data prior to liberalization. Standard errors in parentheses are clustered at the bank level and p-values are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The results are presented in Table 8. A 1 percentage point increase in bank exposure is associated with a 0.30 percentage point larger decline in the probability that a loan is restructured when comparing the post-liberalization period with the pre-liberalization period. Among loans originated only after liberalization, a 1 percentage point increase in exposure is associated with approximately a 0.8 percentage point lower restructuring probability. The estimates are similar when we restrict the sample to newly formed lending relationships.

Taken together with the evidence on the volume of credit in new lending relationships, these results point to changes in the way more-exposed banks make and manage loans. More-exposed banks initially lend cautiously to new borrowers and increase lending as relationships develop. At the same time, their loans are subsequently less likely to be restructured. This pattern is consistent with improvements in screening and monitoring following liberalization, although restructuring decisions may also reflect differences in banks' ex-post loan-management policies.

6.3 Why Did Bank Behavior Change? Foreign Equity versus Foreign Debt

The preceding results raise a further question: why did greater exposure to the liberalization improve banks' lending practices? One possibility is that the form of foreign capital mattered.

Both foreign borrowing and foreign equity can relax a bank's funding constraint, but equity participation potentially operates through an additional channel. Foreign shareholders acquire an ownership stake in the bank and may therefore have incentives to monitor management, influence governance, or facilitate the transfer of information and managerial practices. Foreign debt, by contrast, can expand the resources available to the bank without necessarily generating the same ownership and governance incentives. If the improvement in credit allocation reflected only a relaxation of funding constraints, we might expect foreign borrowing and foreign equity to be similarly related to subsequent lending decisions. If equity participation also changed bank incentives or internal decision-making, the relationship should instead be stronger for foreign equity.

We examine this distinction using a bank-level long-difference exercise. For each bank, we first calculate the change in the share of lending directed to high-MRPK firms:

$$\Delta \textit{High MRPK Lending Share}_b = \overline{\textit{High MRPK Lending Share}}_{b,\textit{Post}} - \overline{\textit{High MRPK Lending Share}}_{b,\textit{Pre}}. \quad (17)$$

This variable captures whether a bank shifted its lending portfolio towards more productive (ex-ante high MRPK) firms following liberalization. We then relate this change to alternative measures of bank's foreign financing in the post-reform period using the following regression specification:

$$\Delta \textit{High MRPK Lending Share}_b = \alpha + \beta \textit{Foreign Financing Measure}_b^{\textit{Post}} + \varepsilon_b, \quad (18)$$

where $\textit{Foreign Financing Measure}_b^{\textit{Post}}$ is alternatively the bank's post-reform foreign equity shareholding, measures capturing the post-reform size of large foreign shareholdings (as a proxy for concentration of foreign ownership) or the post-reform share of foreign borrowing.

Column (1) of Table 9 reports the OLS estimates. Post-reform foreign equity shareholding is positively associated with the reallocation of credit toward productive firms. A 1 percentage point higher foreign equity shareholding in the post-reform period is associated with approximately a 0.34 percentage point larger post-minus-pre increase in the share of lending going to high-MRPK firms. The results are similar when we focus on the presence of large foreign shareholders. A 1 percentage point higher combined shareholding of the three largest foreign investors is associated with a 0.36 percentage point larger increase in the high-MRPK lending share, while the corresponding estimate for the five largest foreign investors is 0.31 percentage points. Banks in which foreign shareholders hold more than 51 percent of equity experience an approximately 11.6 percentage point larger increase in the

share of lending directed to high-MRPK firms.

These OLS relationships are potentially endogenous. Foreign investors may choose banks on the basis of characteristics that are themselves related to subsequent lending decisions and post-reform foreign ownership may respond to changes in bank performance. We therefore complement the OLS exercise with an instrumental-variables specification. Specifically, we instrument the post-reform equity measures using two pre-determined sources of variation: an indicator for whether the bank was included in the MSCI index prior to the reform and the bank's pre-reform exposure measure used throughout the paper. The first stage can then be written as:

$$\textit{Foreign Financing Measure}_b^{Post} = \pi_0 + \pi_1 MSCI_b^{Pre} + \pi_2 Exposure_b + u_b. \tag{19}$$

and the second stage replaces $\textit{Foreign Financing Measure}_b^{Post}$ in equation 18 with its predicted value from this first stage. The motivation is that both variables capture pre-existing differences in banks' access and visibility to foreign equity investors before the reform while determined prior to the post-reform change in lending allocation.

The results from the IV specification are presented in Column 2 of Table 9. The IV estimates reinforce the OLS patterns. The coefficient on post-reform foreign equity shareholding is 0.33, very close to the OLS estimate of 0.34. The IV coefficients are also positive for the shareholdings of the three and five largest foreign investors and for the indicator for foreigners holding more than 51 percent of bank equity. The first-stage F-statistics range from 12.25 to 43.47 across these specifications. The Hansen overidentification test does not reject the overidentifying restrictions in any of the IV specifications.

In contrast, we find no analogous relationship for foreign debt. The coefficient on the post-reform foreign borrowing share is -0.11 with a standard error of 0.21, and is statistically indistinguishable from zero. Thus, banks that relied more heavily on foreign borrowing did not experience a systematically larger shift in lending toward high-MRPK firms.

We interpret this exercise cautiously. The bank-level analysis is based on only 32 banks and post-reform foreign ownership is not randomly assigned. The IV estimates address one source of endogeneity by using predetermined measures of foreign-investor access, but their causal interpretation requires the additional exclusion restriction that pre-reform MSCI inclusion and bank exposure affect the subsequent change in credit allocation through their effect on post-reform foreign equity participation only, rather than through an independent channel. The failure to reject the Hansen test is consistent with, but does not establish, this assumption. Accordingly, we view these estimates as supporting evidence on the mechanism rather than as a separate causal decomposition of the reform effect. Their value is that they line up with the broader set of results in the paper. The banks that end up with greater foreign equity participation — and particularly larger foreign shareholder

TABLE 9. Decomposing Channels: Foreign Equity vs. Foreign Debt

Dependent variable: Change (Post – Pre) in Share of Lending to More Productive Firms		
	(1) OLS	(2) IV
Panel A: Measure of Foreign Equity Holdings		
Post-Reform Foreign Equity Shareholding	0.344**	0.329**
	(0.154)	(0.156)
R-squared	0.151	0.151
F-stat (First Stage)		43.47
Panel B: Measures of Concentration of Foreign Equity Holdings		
Shareholding of Top 3 Foreign Entities	0.360**	0.599*
	(0.163)	(0.321)
R-squared	0.091	0.051
F-stat (First Stage)		13.84
Shareholding of Top 5 Foreign Entities	0.313**	0.530*
	(0.153)	(0.292)
R-squared	0.092	0.048
F-stat (First Stage)		12.25
Indicator for Majority Foreign Shareholding (>51%)	11.611**	19.769**
	(5.408)	(9.779)
R-squared	0.065	0.033
F-stat (First Stage)		15.54
Panel C: Measure of Foreign Debt Shares		
Post-Reform Foreign Borrowing Share	-0.108	
	(0.211)	
R-squared	0.017	
Observations	32	32

Notes: This table presents the results from equation 18. The dependent variable is the change (post – pre) in the share of bank lending to high-MRPK firms (given by equation 17). Column 1 gives the results from a simple OLS specification. Column 2 controls for endogeneity of post-reform equity flows by using pre-reform MSCI index membership and the bank exposure measure as instruments for post-reform foreign equity holdings and concentration. The Hansen J overidentification test fails to reject in every IV specification. Robust standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

positions — are the banks that shift their lending more toward productive firms. The same pattern is not present for foreign borrowing.

Taken together, the evidence suggests that the consequences of liberalization depended not only on the amount of foreign capital that banks received, but also on the form through which that capital entered the banking system. Foreign borrowing could relax funding constraints, but foreign equity participation also brought foreign investors inside the ownership structure of the bank. The accompanying increases in human-capital investment, more cautious lending to new borrowers, and lower restructuring rates are consistent with foreign equity affecting banks' screening, monitoring, and internal decision-making. While these results do not establish a unique causal mechanism, they point to equity participation as an important channel through which liberalization improved the allocation of credit.

In this sense, the form of foreign capital appears to matter: the improvement in lending quality following liberalization is associated with foreign equity participation rather than simply with greater access to foreign funding.

7. Conclusion

In this paper, we study how foreign capital liberalization affects the real economy through the domestic banking system. We exploit a reform that increased foreign shareholding limits in Indian banks and trace its effects from bank funding to credit allocation, firm outcomes and aggregate productivity. We find that banks with greater exposure to the liberalization see a reduction in their funding constraints and expand credit. Importantly, this additional credit is directed disproportionately towards firms with higher marginal returns to capital. These firms subsequently expand, the dispersion in the marginal revenue product of capital across firms declines, and this leads to an improvement in aggregate productivity.

These findings speak to the broader debate on the impact of capital flows and international financial integration. A large literature links episodes of capital inflows to rapid credit growth, rising financial vulnerability and, in some cases, financial crises. The episode we study looks different. Despite a substantial expansion in lending, we find an improvement rather than a deterioration in the allocation of credit. Our mechanism results suggest that the response went beyond a simple relaxation of banks' funding constraints: more-exposed banks invested more in human capital, lent more cautiously when forming new relationships, and experienced lower rates of loan restructuring.

The form of foreign capital appears to have played an important role. Banks with greater post-reform foreign equity participation and with larger foreign shareholder positions experienced larger shifts in lending toward more productive firms, whereas there is no such analogous relationship for foreign borrowing. While this evidence is supportive rather than

fully causal, it is consistent with foreign equity affecting not only the resources available to banks but also their incentives, governance, and internal decision-making.

Taken together, the results suggest that the effects of international financial integration depend not only on the amount of capital that enters the domestic financial system, but also on the channel through which it enters. Greater access to international capital need not come at the cost of poorer credit allocation when foreign participation strengthens financial intermediaries' incentives and capacity to allocate those resources effectively. However, it bears emphasizing that, in this paper, we have looked at only one aspect of allowing foreign capital into the banking sector. Before drawing definitive policy conclusions, there are other considerations to take into account. For instance, during a crisis in the foreign investors' own country, the investor might pull financing, precipitating a banking crisis in the home country — possibly requiring the home-regulator to ring-fence assets. Nevertheless, by documenting how foreign equity participation in domestic banks reshapes credit allocation and firm-level outcomes, our paper highlights an important channel through which international financial integration can contribute to productivity and growth.

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Appendix

A Data Appendix

TABLE A1. Summary Statistics

Variable Name	Units	Mean	Median	SD
Panel A: Bank Characteristics				
Total Assets	INR Billions	495.41	233.46	827.86
Total Advances	INR Billions	250.47	106.48	435.25
Total Deposits	% Liabilities	83.13	85.45	8.78
Total Borrowings	% Liabilities	4.04	2.01	6.86
Total Foreign Borrowings	% Borrowings	35.70	24.02	33.26
Total Foreign Shareholding	%	14.23	6.11	18.50
Panel B: Firm Characteristics				
Total Assets	INR Millions	2757.07	271.30	20256.43
Total Sales	INR Millions	2435.13	298.30	26258.79
Total Bank Borrowings	INR Millions	466.08	62.90	2975.85
No. of Banking Relations	No.	3.16	2.00	3.04

Note: This table reports summary statistics (1998-2008 averages) for firms and banks used in the analysis. Bank balance-sheet data is from the Reserve Bank of India and firm balance-sheet data is from CMIE Prowessdx. The number of banking relationships for each firm are calculated from the MCA loan-level data.

A.1 Data construction: Loan-level data

We follow the existing literature that uses loan-level data (formally referred to as "Index of Charges") from the MCA to construct a loan-level panel matched with firm and bank level characteristics (Kulkarni (2020)). The MCA maintains records of all secured loans on which a charge has been registered by the lender under the Companies Act, 2013. Under this Act, lenders may register a charge on secured loans. If a loan is not registered, then in the event of bankruptcy or liquidation, it is treated on par with an unsecured loan. Given these material consequences, most lenders do register their charges with the MCA (Chopra et al. (2021)).

To construct our dataset, we begin with the Company Identification Numbers (CINs) for all firms (46,069 firms) in the ProwessDx database and collect data on charges for each of these firms from the MCA website. Not all firms in ProwessDx have a corresponding entry in the MCA database. We are able to obtain charge data for 30,586 firms (last downloaded in May 2023). Each loan record between a bank and a firm in the MCA database is associated with a unique identifier (Charge ID). The database provides details on the lending institution, the firm name and ID, the loan amount , the date on which the loan was issued, the dates of modification and the date of satisfaction or repayment of the loan.

We winsorize the data to address extreme outliers and remove duplicate loan entries. Since the dataset records both loan origination and repayment dates, we can compute the outstanding credit between each bank-firm pair. We then construct a balanced panel of outstanding credit at the bank-firm-year level. This loan-level panel is matched with firm

and bank level characteristics which forms the basis of our main analysis.

A.2 Data construction: FII flows at the bank level

We construct a dataset on FII inflows to each bank using FII equity trade data from NSDL. We use this flow data to demonstrate that the bank exposure measure serves as a valid instrument for predicting equity inflows after liberalization. NSDL provides data on equity trades starting from 2003. The lack of data before 2003 is not a concern, as we require flow data only for the post-liberalization years.

We download the FII equity trade data from NSDL for the period 2003–2008 and filter it to include only the banks in our sample. Using this data, we construct series on FII inflows, FII outflows and net FII inflows. To calculate FII inflows, we aggregate the values of secondary market purchases, primary market purchases, rights issue purchases and preferential allotments. FII outflows are calculated using the values of secondary market sales for each bank. Finally, net FII inflows are computed as the difference between total FII inflows and total FII outflows.

A.3 Data construction: MRPK

To construct firm level MRPK's, we follow the method used by Bau and Matray (2023) and use the deflator data that they provide. Assume that firms follow a Cobb-Douglas revenue production function:

$$Revenue_{ijt} = TFPR_{ijt} K_{ijt}^{\alpha_j^k} L_{ijt}^{\alpha_j^l} M_{ijt}^{\alpha_j^m} \tag{A1}$$

where i denotes a firm, j denotes an industry and t denotes a year. $Revenue_{ijt}, K_{ijt}, L_{ijt}, M_{ijt}$ are measures of sales, capital, the wage bill and materials and $TFPR_{ijt}$ is the firm-specific unobserved revenue productivity.

In this scenario, $MRPK = \frac{\partial Revenue_{i,t}}{\partial K_{i,t}} = \alpha_j^k \frac{\partial Revenue_{i,t}}{\partial K_{i,t}}$. Thus, $\frac{Revenue_{i,t}}{K_{i,t}}$ provides a within-industry measure of MRPK, under the assumption that all firms in an industry share the same α_j^k. We therefore calculate MRPK as the ratio of $\frac{Sales}{GrossFixedAssets}$ for each firm. We deflate sales using the GDP deflator and Gross Fixed Assets using the Capital Deflator.

B Threats to Identification

A central threat to identification is that banks with higher pre-liberalization foreign shareholding may simply be different kinds of banks – larger, better capitalized or better managed – and may therefore have already been on differential trajectories in credit allocation even absent the reform. If so, the difference-in-differences estimates would conflate the causal effect of liberalization with these pre-existing differences. We address this threat in three ways. First, Figure B1A tests whether a broad set of ex-ante bank characteristics including size, the loan-deposit ratio, the capital-asset ratio, profitability (return on assets and net interest margin), asset quality (the gross NPA ratio and the loan modification rate) are systematically related to the exposure measure. Each coefficient plotted is from a separate regression of a standardized pre-reform characteristic on the exposure measure. Most characteristics show no significant relationship with exposure; the coefficient on foreign liabilities share is significant at the 10% level ($p = 0.058$) but not at the 5% level and overall the results indicate that more exposed banks are not observably different from less exposed banks along these dimensions. Since our key outcome of interest is bank lending, we also examine whether the bank exposure measure was systematically related to their pre-reform lending behavior. Figure 3 shows the scatter plot between the bank exposure measure and their pre-reform overall lending (Figure 3A) and their pre-reform credit allocation behavior – i.e. lending to more vs. less productive firms (Figure 3B). There is no significant relationship between bank exposure and pre-reform lending and credit allocation behavior. Second, and more directly relevant to the identifying assumption, the dynamic event-study specifications show no statistically significant differential pre-trends between more and less exposed banks in the years prior to the reform, indicating that more exposed and less exposed banks were on similar trajectories before liberalization took effect. Third, all specifications include bank fixed effects and control for ex-ante bank characteristics (specifically, bank size and liquidity) interacted with the post-liberalization indicator, directly absorbing any differential trends associated with these observable characteristics rather than relying solely on the balance tests and pre-trends to rule them out.

A second potential concern is that India's banking sector experienced an earlier, related policy change in 2001, when the FDI cap in private banks was raised to 49%. If this earlier reform had a meaningful effect on bank behavior – particularly if it affected more exposed and less exposed banks differentially – our pre-period would not represent a clean counterfactual and any differences we attribute to the 2004 reform could instead reflect delayed or ongoing effects of the 2001 change. However, as discussed in the institutional background section, the 2001 reform had little practical effect on foreign shareholding in Indian banks, largely due to regulatory ambiguity over whether the 49% cap applied to FDI alone or to total foreign shareholding and confusion arising from uncoordinated signals across the Ministry of Finance, the Reserve Bank of India and the Ministry of Commerce. Figure 1 confirms

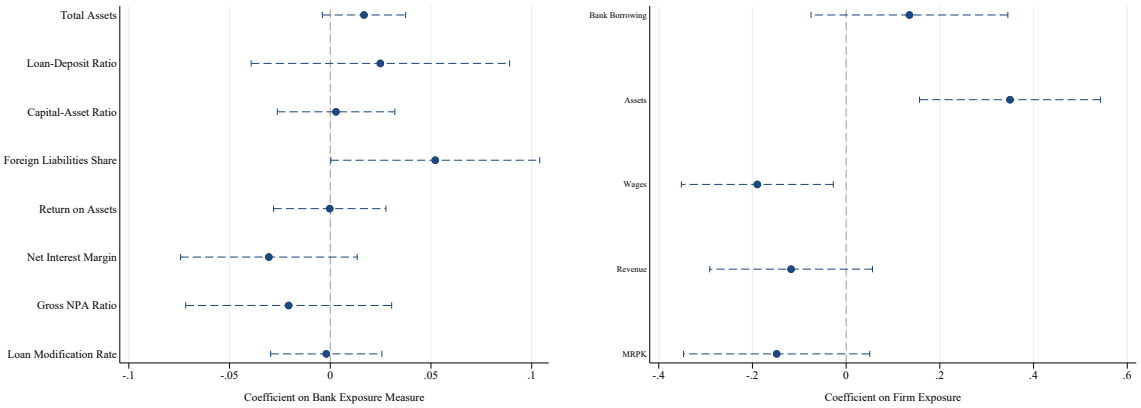
that foreign shareholding in Indian banks did not respond meaningfully to the 2001 change. Consistent with this, the dynamic event-study specifications show no differential pre-trends between more and less exposed banks across the pre-period spanning the 2001 reform, indicating that it did not generate differential effects across banks that could confound the 2004 liberalization estimates.

A third concern is that public and private banks in India differ fundamentally in ownership structure and governance. We include both in the analysis because public banks also experienced meaningful foreign capital inflows over this period, moving from near-negligible foreign shareholding toward their own regulatory ceiling, making them a relevant part of the credit market rather than a group that can simply be excluded. This means public banks cannot be treated as a clean, untreated control group in the traditional sense. We instead address the concern that ownership differences drive the results directly: all specifications include bank fixed effects, which absorb any time-invariant differences between public and private banks, and the dynamic event-study specifications show no differential pre-trends across banks in the pre-liberalization period, indicating that these ownership-related differences are not generating differential trajectories prior to the reform. The fact that public banks also received capital inflows over this period does not itself threaten identification, since our design does not rely on public banks serving as an untreated benchmark – it relies on continuous variation in ex-ante exposure and the absence of differential pre-trends across that variation.

A further concern is that India experienced a broader capital inflow and credit boom over this period and the observed relationship between exposure and lending could simply reflect this general macroeconomic environment rather than the 2004 reform specifically. To address this, we include year fixed effects in all specifications which absorb any shock common to all banks in a given year – including a general rise in capital inflows or credit supply across the Indian banking sector. The relationship we estimate is therefore identified only from variation in how banks with different levels of ex-ante exposure respond differently within the same year, not from the aggregate level of lending or capital inflows in any given year. Bank fixed effects further absorb any time-invariant differences across banks, so the estimated coefficient captures how a bank's lending changes over time as a function of its exposure, relative to other banks controlling for common shocks across the banking sector.

Lastly, for the firm-level analysis, we construct a firm-specific exposure measure as a weighted average of the exposure of the banks each firm borrows from prior to liberalization. This allows us to study how firm outcomes evolve differentially depending on a firm's exposure to bank liberalization through its lending relationships. A natural concern is that firms with higher exposure – i.e. firms that happen to borrow from more exposed banks – might be systematically different from less exposed firms, and these differences,

rather than the reform could be driving the firm-level results. We test this using a firm-level balance test, regressing firm exposure on pre-reform firm characteristics. Figure B1B shows the standardized coefficients from this test: firm size and wages paid (as a share of assets) are significantly related to exposure, indicating that larger firms tend to borrow from more exposed banks. The other characteristics shown – including secured bank borrowing, the sales-to-assets ratio and, most importantly, pre-reform productivity (MRPK) show no significant relationship with exposure. Since our main specifications include firm fixed effects, any fixed differences in firm size and wages that are correlated with exposure are absorbed. The absence of a pre-existing relationship between exposure and productivity is particularly reassuring and validates the key results of credit allocation and productivity.



(A) Bank Exposure (B) Firm Exposure

FIGURE B1. Balance Tests: Exposure and Ex-Ante Characteristics

Notes:

(a) The figure in Panel A examines whether the exposure measure is systematically related to banks' pre-reform characteristics. Each coefficient is from a separate OLS regression of the standardized pre-reform bank characteristic on the bank exposure measure, with heteroskedasticity-robust SE and 95% confidence intervals shown. Most characteristics are not significantly related to exposure, with the exception of the foreign liabilities share, which is weakly significant at the 10% level.

(b) The figure in Panel B examines whether the firm exposure measure is systematically related to firms' pre-reform characteristics. Bank Borrowing, Assets, and MRPK are in natural logarithms; Wages and Revenue are expressed as a share of total assets. Each coefficient is from a separate OLS regression of the standardized pre-reform firm characteristic on the firm exposure measure, with heteroskedasticity-robust SE and 95% confidence intervals shown.

C Bank Lending: Additional Results

Volume of Credit in New Lending Relationships. In Section 4, we show that more exposed banks were more likely to form new lending relationships following the liberalization, particularly with high-MRPK firms. In this section, we look more closely at lending within these newly formed relationships. We first examine whether bank exposure affects the size of the initial loan and then study how credit evolves over the first three years of the relationship. This analysis complements the extensive margin entry likelihood results by showing how banks extend credit once a new lending relationship is formed.

We first examine whether more exposed banks make larger or smaller initial loans to new borrowers. We restrict the sample to firm-bank pairs that form a new lending relationship in the post-liberalization period and estimate:

$$\ln(\textit{Loan Amount}_{f,b}) = \beta_1 Exposure_b + X_b'\theta + \alpha_f + \epsilon_{f,b}. \tag{C1}$$

The dependent variable is the size of the first loan made by bank b to firm f. $Exposure_b$ is the bank exposure measure defined earlier and X_b includes ex-ante bank characteristics such as bank size and liquidity. Firm fixed effects, α_f, allow us to compare the size of initial loans made by differently exposed banks to the same firm. We also estimate a specification replacing firm fixed effects with region-sector fixed effects. Standard errors are clustered at the bank level and p-values are obtained using wild cluster bootstrap inference.

Table C1 (Panel A) reports the results. More exposed banks make significantly smaller first loans to new borrowers. A 1 percentage point increase in bank exposure is associated with a smaller first-loan size by 3.6% in the firm fixed effects specification and by 3.2% in the region-sector fixed effects specification. These results suggest that more exposed banks initially extend smaller amounts of credit when forming a new lending relationship.

We next examine whether this effect differs across firm types. We estimate:

$$\ln(\textit{Loan Amount}_{f,b}) = \sum_d \beta_d D_f^d Exposure_b + X_b'\theta + \alpha_f + \epsilon_{f,b}, \tag{C2}$$

where D_f^d denotes the firm categories defined in the main text.

Table C1 reports the results. Panel B examines heterogeneity by firm productivity. The decline in first-loan size is very similar for high- and low-MRPK firms. In the firm fixed effects specification, a 1 percentage point increase in exposure is associated with a smaller first-loan size by approximately 3.2% for both groups. The corresponding estimates in the region-sector fixed effects specification are 2.9% for high-MRPK firms and 2.8% for low-MRPK firms. Although the high-low differential is statistically significant, it is small in magnitude, indicating little economically meaningful heterogeneity by productivity.

Panel C examines heterogeneity jointly by productivity and collateral. The decline in first-loan size is present across most firm categories. It is somewhat larger for high-MRPK, low-FA firms than for high-MRPK, high-FA firms. The difference between these two groups is statistically significant in the region-sector fixed effects specification but not in the firm fixed effects specification. Overall, the results suggest that the tendency of more exposed banks to make smaller initial loans is fairly broad across firm types.

Finally, we examine how lending evolves over the first three years of a new lending relationship. We first estimate the following specification using relationships formed in the post-liberalization period:

$$\ln(\textit{Credit}_{f,b,t}) = \sum_{k=1}^{3} \delta_k \left(Year_k \times Exposure_b\right) + \alpha_{f,t} + \gamma_{f,b} + \epsilon_{f,b,t}, \tag{C3}$$

where $Year_k$ indicates the first, second or third year of the lending relationship. The specification includes firm-time fixed effects and firm-bank fixed effects.

We also estimate a difference-in-differences specification using relationships formed before the liberalization as a comparison group:

$$\ln(\textit{Credit}_{f,b,t}) = \sum_{k=1}^{3} \delta_k \left(Year_k \times Exposure_b\right) + \sum_{k=1}^{3} \beta_k \left(Year_k \times Exposure_b \times Post_t\right) + \alpha_{f,t} + \gamma_{f,b} + \epsilon_{f,b,t} \tag{C4}$$

In both specifications, standard errors are clustered at the bank level and p-values are obtained using wild cluster bootstrap inference. The coefficients capture how credit extended by more exposed banks changes as a new lending relationship develops.

Table C2 reports the results. In both specifications, the Year 2 coefficient is positive and statistically significant. A 1 percentage point increase in exposure is associated with a 0.62% increase in credit in the second year of the relationship in the post-reform-only specification and a 0.66% increase in the difference-in-differences specification. The estimates for Years 1 and 3 are positive but not statistically significant.

Taken together, these results show that more exposed banks make smaller initial loans when forming new lending relationships, with relatively little variation in this pattern across firm types. We also find some evidence that lending increases as these relationships develop. This pattern is consistent with banks screening new borrowers cautiously at the start of a relationship and expanding credit as they learn more about borrower quality.

TABLE C1. Bank Exposure to Liberalization and First-Loan Size at Entry

Dependent variable: Value of First Loan		
	(1)	(2)
Panel A: Average Effect		
Exposure	-0.036***	-0.032***
	(0.008)	(0.007)
Observations	5,649	9,418
Adj. R-squared	0.394	0.139
Panel B: MRPK		
Exposure × High MRPK	-0.032***	-0.029**
	(0.009)	(0.007)
Exposure × Low MRPK	-0.032***	-0.028*
	(0.010)	(0.007)
High vs Low Differential (t-test)	-0.001*	-0.001*
	(0.006)	(0.002)
Observations	2,410	2,989
Adj. R-squared	0.402	0.164
Panel C: Joint MRPK and FA		
Exposure × High MRPK-High FA	-0.029**	-0.021
	(0.009)	(0.008)
Exposure × High MRPK-Low FA	-0.037***	-0.039*
	(0.011)	(0.007)
Exposure × Low MRPK-High FA	-0.031***	-0.019
	(0.010)	(0.008)
Exposure × Low MRPK-Low FA	-0.036	-0.068***
	(0.017)	(0.017)
HH vs HL Differential (t-test)	0.008	0.018*
	(0.005)	(0.005)
Observations	2,408	2,985
Adj. R-squared	0.402	0.177
Firm FE	Yes	No
Region-Sector FE	No	Yes

Note: The table reports the coefficients of equations C1 and C2. It shows the impact of bank exposure to liberalization on the volume of credit (in logarithms) to a firm conditional on the firm and bank starting a new lending relationship in the post-liberalization period (Panel A), and according to the type of firm (Panels B–C). Panel B compares firms with high versus low productivity (measured by the ex-ante marginal revenue product of capital (MRPK)). Panel C decomposes jointly by productivity and collateral (measured by ex-ante fixed assets (FA)). The reported differential (HH vs HL) is the collateral premium among high-MRPK firms. Pre-period bank-level controls (assets, loan-deposit ratio) are included in every panel. Column 1 uses firm fixed effects and column 2 uses region-sector fixed effects. Standard errors in parentheses are clustered at the bank level and p-values are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE C2. Bank Exposure to Liberalization and Credit Evolution After Entry

Dependent variable: $\ln$(Outstanding Credit)		
	(1) Post-Reform Only	(2) DiD
Year 1 $\times$ Exposure (β_1)	0.0035	-0.0001
	(0.0012)	(0.0015)
Year 2 $\times$ Exposure (β_1)	0.0062***	-0.0004***
	(0.0013)	(0.0019)
Year 3 $\times$ Exposure (β_1)	0.0047	0.0004
	(0.0022)	(0.0025)
Year 1 $\times$ Exposure $\times$ Post (β_2)		0.0036
		(0.0013)
Year 2 $\times$ Exposure $\times$ Post (β_2)		0.0066***
		(0.0019)
Year 3 $\times$ Exposure $\times$ Post (β_2)		0.0043
		(0.0033)
Firm-Year FE	Yes	Yes
Firm-Bank FE	Yes	Yes
Observations	9,846	13,744
Adj. R-squared	0.9333	0.9420

Note: The table reports the coefficients of equations C3 and C4. It shows the impact of bank exposure to liberalization on the evolution of credit over the first three years of a new firm-bank lending relationship. Column 1 restricts the sample to relationships formed after the reform (first loan 2005–2008) and estimates the year-by-year exposure gradient directly. Column 2 adds a pre-reform comparison cohort (first loan 1998–2001, chosen so that years 1–3 of these relationships fall entirely before 2004 and are therefore unaffected by the reform) and estimates this gradient as a difference-in-differences specification. A rising sequence of coefficients (Year 1 to Year 3) is consistent with exposed banks screening new borrowers cautiously at the start of a relationship and scaling up lending as information about firm creditworthiness accumulates. Both columns include firm-year and firm-bank fixed effects. Standard errors in parentheses are clustered at the bank level and p-values are from a wild cluster bootstrap t procedure. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Aggregate Productivity Effects

In this section, we use an alternate method to assess the aggregate effect of liberalization induced capital reallocation on aggregate productivity using the sufficient-statistics approach proposed by Sraer and Thesmar (2023).

Sraer and Thesmar (2023) show that under the assumption of log-normality of MRPK, the aggregate change in total factor productivity (TFP) due to a reform depends on the D-i-D estimates of the effect of the reform on (i) changes in the variance of $ln(MRPK)$ within industries, (ii) changes in the mean of $ln(MRPK)$ by industry and (iii) changes in the covariance of $ln(MRPK)$ and $ln(Sales)$ by industry. The aggregate change in TFP can then be expressed by the following formula:

$$\begin{aligned}\Delta \log(\text{TFP}) = &-\frac{\alpha}{2}\left[1+\frac{\alpha\theta}{1-\theta}\right]\sum_{s=1}^{S}\kappa_s\,\widehat{\Delta\Delta\sigma^2(s)} \\ &-\frac{\alpha}{2}\left[1+\frac{\alpha\theta}{1-\theta}\right]\sum_{s=1}^{S}(\phi_s-\kappa_s)\left(\widehat{\Delta\Delta\mu(s)}+\widehat{\Delta\Delta\sigma_{\text{MRPK,py}}(s)}+\frac{1}{2}\frac{\alpha\theta}{1-\theta}\widehat{\Delta\Delta\sigma^2(s)}\right)\end{aligned} \tag{D1}$$

Specifically, $\widehat{\Delta\Delta\sigma^2(s)}$ is the D-i-D estimate of the reform on within-industry var(ln(MRPK)), $\widehat{\Delta\Delta\mu(s)}$ is the D-i-D estimate of the reform on industry's mean(ln(MRPK)) and $\widehat{\Delta\Delta\sigma_{MRPK,py}(s)}$ is the D-i-D estimate of the reform on industry's cov(ln(MRPK),ln(Sales)). Further, κ_s is the capital share of industry s and ϕ_s is the sales share of industry s before the shock, α is the share of capital in firms' production functions and θ denotes the elasticity of substitution across varieties. The first term of this equation captures the effect of the reform on misallocation within an industrial sector, whereas, the second term captures the effect of the reform on misallocation between sectors.

Following their methodology, we first verify that the log-normality assumption underlying the aggregation formula holds in our sample. We restrict the sample to a balanced panel of firms observed in every year of the sample period and winsorize $log(MRPK)$ and $log(Sales)$ each year to limit the influence of outliers. Figure D1 presents the normal probability plot of standardized log-MRPK for all firms in our sample and confirms that the log-normality assumption holds.

We next estimate the effect of the banking liberalization on three sufficient statistics of the log-MRPK distribution at the sector (NIC-2-digit) level: the variance of log-MRPK, the mean of log-MRPK and the covariance of log-MRPK with log sales. For each sector-year cell, these moments are computed only when the cell contains at least five firms to ensure they are estimated with reasonable precision. We define sector-level exposure to the reform, $Exposure_s$ as the average of firms' individual exposure measures within sector s.

We then estimate the following difference-in-differences regression:

$$M_{s,t} = \delta_t + \eta_s + \beta_M Exposure_s \times Post_t + \mu_s \times t + \epsilon_{s,t} \tag{D2}$$

where $M_{s,t}$ denotes, in turn, var(ln(MRPK)), mean(ln(MRPK)) or cov(ln(MRPK),ln(Sales)) for sector *s* in year *t*. $Post_t$ is a dummy variable equal to one for the post-liberalization period. η_s and δ_t are sector and year fixed effects respectively and $\mu_s \times t$ are sector specific trends. Standard errors are clustered at the sector level.

We estimate this equation under three alternative specifications of treatment intensity. In the first, we use the continuous exposure measure $Exposure_s$ directly so that β_M captures the impact of a 1 percentage point increase in sector exposure on each moment and the total treatment effect for a given sector is obtained by scaling β_M by that sector's own exposure value. In the second, we discretize exposure into a binary indicator using the sample median, splitting sectors into high- and low-exposure groups. Here, β_M is interpreted as the differential effect of the reform on high-exposure relative to low-exposure sectors, under the identifying assumption that low-exposure sectors are only minimally affected by the reform. In the third, we replace the continuous measure with a set of indicator variables for exposure quartiles, omitting the lowest-exposure quartile as the reference category. This specification relaxes the linearity imposed by the continuous measure while still allowing the treatment effect to vary across the exposure distribution.

The results for these estimations are presented in Table D1. For $Var(ln(MRPK))$, the continuous specification shows a negative and statistically significant coefficient indicating that more exposed sectors experienced a decline in within-sector capital misallocation following liberalization. This pattern is corroborated by the median-split specification, where high-exposure sectors see a significantly larger decline in dispersion relative to low-exposure sectors.The quartile specification shows a broadly monotonic pattern — the decline in dispersion grows in magnitude moving from Q2 to Q4 — though only the Q4 coefficient is individually significant.

Figure D2 shows the corresponding dynamic D-i-D estimates for $Var(lnMRPK)$ using the continuous sector-level exposure measure specification. There are no significant pre-trends — giving support to the identifying assumption that more- and less-exposed sectors were not on differential trajectories prior to liberalization. Following the reform, the coefficients decline steadily and become more negative in each subsequent year, consistent with a persistent, growing decline in within-sector capital misallocation in more exposed sectors over time.[25]

For $Mean(ln(MRPK))$ and $Covariance(ln(MRPK), ln(Sales))$, the coefficients are

[25]This sector-level pattern is consistent with and lends aggregate-level support to the firm-level heterogeneity results discussed in the previous section which similarly point to a decline in misallocation following the reform.

not statistically significant across the various specifications. However, the consistent signs on the coefficients are informative – the positive coefficients on mean log-MRPK across all three specifications suggest a mild increase in the average capital wedge in more exposed sectors, while the uniformly negative coefficients on the covariance term suggest that the reform weakened the association between productivity and capital wedges — i.e., more productive firms in exposed sectors became relatively less distorted, consistent with capital reallocating toward higher-productivity firms.

For the aggregation exercise, we use the coefficients from the median-split specification reported in column (2). We use standard calibration parameters from the literature (Sraer and Thesmar (2023); David and Venkateswaran (2019)) and set $\alpha = 0.33$ and $\theta = 0.83$, corresponding to a price elasticity of demand of 6. Finally, we calculate κ_s and ϕ_s for each sector using the pre-liberalization shares of capital (measured using real gross fixed assets) and output (measured using real sales), respectively. The change is aggregate TFP is then:

$$\begin{aligned}\Delta \log(TFP) \approx & \underbrace{-\frac{\alpha}{2}\left(1+\frac{\alpha\theta}{1-\theta}\right)\sum_{s=1}^{S}\kappa_s \Delta\widehat{\Delta\sigma^2(s)}}_{\text{Within-industry reallocation: } 0.0274} \\ & + \underbrace{-\frac{\alpha}{2}\left(1+\frac{\alpha\theta}{1-\theta}\right)\sum_{s=1}^{S}(\phi_s-\kappa_s)\left[\widehat{\Delta\Delta\mu(s)}+\widehat{\Delta\Delta\sigma_{MRPK,py}}(s)+\frac{1}{2}\frac{\alpha\theta}{1-\theta}\widehat{\Delta\Delta\sigma^2(s)}\right]}_{\text{Cross-industry reallocation: } 0.0092} \\ \approx & \ 0.0366. \end{aligned} \tag{D3}$$

The decomposition in equation (D3) implies that most of the aggregate productivity gain comes from reallocation within industries. The within-industry component contributes 0.0274 log points, reflecting the larger decline in within-sector dispersion in $log(MRPK)$ among more-exposed sectors (relative to less-exposed sectors). This is consistent with an improvement in the allocation of capital across firms within sectors. The cross-industry component contributes a further 0.0092 log points, indicating that changes in the allocation of capital across sectors modestly reinforce the within-sector gains. Together, the two terms imply an increase in aggregate TFP of 0.0366 log points, corresponding to approximately 3.73% in exact percentage terms.

This aggregation relies on the identifying assumption that below-median-exposure industries were not directly affected by the reform, so that they provide the counterfactual evolution of MRPK moments in the absence of liberalization. Under this assumption, the 3.7% estimate can be interpreted as the aggregate TFP effect of liberalization. If below-median-exposure industries were also affected, the estimate instead captures the additional

productivity improvement in above-median-exposure industries relative to below-median-exposure industries.

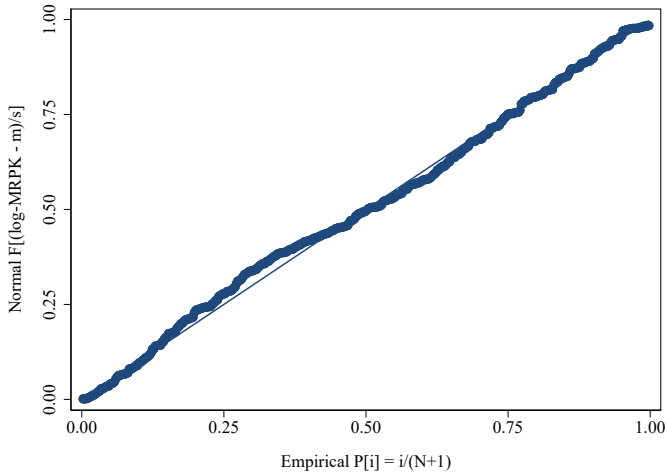


FIGURE D1. Log-Normality of MRPK

Note: This figure shows the normal probability plot for the distribution of log-MRPK for the firms in our sample.

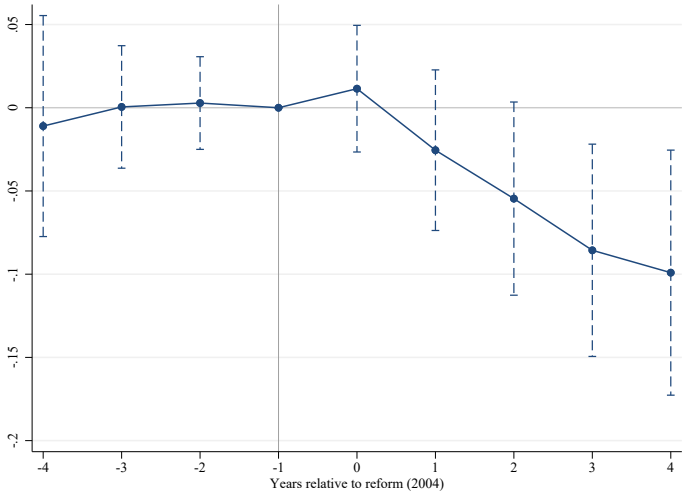


FIGURE D2. Event Study Graph for the Effect of Liberalization on Var(MRPK) Across Firms

Note: This figure reports event study estimates of the effect of liberalization on the variance of log(MRPK). Each point is the coefficient on the interaction between sector-level expsoure and year relative to liberalization, normalized to zero in the year immediately preceding the reform. The confidence intervals are at the 95% level.

TABLE D1. Sector-Level Effects of Reform Exposure on Sufficient Statistics

	(1) Continuous	(2) Median Split	(3) Quartiles
Panel A: Var(ln MRPK)			
Exposure × Post	-0.070***		
	(0.021)		
High Exposure × Post		-0.178*	
		(0.098)	
Q2 Exposure × Post			-0.126
			(0.141)
Q3 Exposure × Post			-0.166
			(0.126)
Q4 Exposure × Post			-0.318**
			(0.156)
Observations	418	418	418
Adj. R-squared	0.898	0.895	0.896
Panel B: Mean(ln MRPK)			
Exposure × Post	0.011		
	(0.009)		
High Exposure × Post		0.039	
		(0.051)	
Q2 Exposure × Post			0.066
			(0.081)
Q3 Exposure × Post			0.088
			(0.073)
Q4 Exposure × Post			0.051
			(0.071)
Observations	418	418	418
Adj. R-squared	0.975	0.975	0.975
Panel C: Cov(ln MRPK, ln Sales)			
Exposure × Post	-0.059		
	(0.035)		
High Exposure × Post		-0.127	
		(0.134)	
Q2 Exposure × Post			-0.200
			(0.208)
Q3 Exposure × Post			-0.231
			(0.154)
Q4 Exposure × Post			-0.212
			(0.211)
Observations	418	418	418
Adj. R-squared	0.915	0.914	0.914
Sector FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Sector-Specific Trends	Yes	Yes	Yes

Note: The table reports difference-in-differences estimates of the effect of reform exposure on sector-level moments of the log-MRPK distribution. Column (1) uses the continuous sector-level exposure measure interacted with the post-reform indicator. Column (2) uses a binary indicator for above-median exposure. Column (3) uses indicators for exposure quartiles, with the lowest-exposure quartile (Q1) as the omitted category. All regressions include sector and year fixed effects, sector-specific linear trends, and cluster standard errors at the sector level (in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.